

**Vidya Vikas Institute of Engineering and Technology,
Mysore
Library and Information Centre**



VTU Question Papers

“INFORMATION SCIENCE & ENGINEERING”

Odd Semester

2022-23

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

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CBCS SCHEME

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18CS51

Fifth Semester B.E./B.Tech. Degree Examination, Jan./Feb. 2023 Management and Entrepreneurship for IT Industry

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Management? List the functional areas of management and explain any two in detail. (10 Marks)
- b. Explain in detail levels of management. (06 Marks)
- c. Briefly describe importance of Staffing. (04 Marks)

OR

- 2 a. Discuss the importance of planning Briefly explain the general steps involved in planning. (10 Marks)
- b. Explain any two Management Approaches. (06 Marks)
- c. Write a difference between Formal and Informal organization. (04 Marks)

Module-2

- 3 a. What is Motivation? Explain Maslow's Need Hierarchy theory of Motivation. (10 Marks)
- b. Differentiate between co-ordination and co-operation. (06 Marks)
- c. Write a difference between Manager and a Leader. (04 Marks)

OR

- 4 a. Define Control. Briefly explain the methods of establishing control. (10 Marks)
- b. What are the merits and demerits of Herzberg's two factor theory? (06 Marks)
- c. List the characteristics of Direction. (04 Marks)

Module-3

- 5 a. Define Entrepreneur. Explain the functions of entrepreneur. (10 Marks)
- b. Briefly describe the stages in entrepreneurial process. (06 Marks)
- c. Write the differences between Entrepreneurs and Intrapreneurs. (04 Marks)

OR

- 6 a. Explain different types of entrepreneurs along with classifications. (10 Marks)
- b. What are the barriers of an entrepreneur? (06 Marks)
- c. Write the differences between Entrepreneur and Manager. (04 Marks)

Module-4

- 7 a. Define the meaning of Project. Explain in detail the various ways of Project Identification. (10 Marks)
- b. Briefly describe the contents of Project Report. (06 Marks)
- c. List the different sequential stages for formulation of Project Report. (04 Marks)

OR

- 8 a. Define ERP? Explain the importance and need of a ERP. (10 Marks)
b. Write short notes on functional areas of management of ERP. (06 Marks)
c. List the planning commission guidelines for preparing industrial projects. (04 Marks)

Module-5

- 9 a. Explain the steps involved in establishing Micro and Small enterprise. (10 Marks)
b. Discuss the case study of air Deccan (Captain G.R. Gopinath). (06 Marks)
c. What is Patent? List different types of Patent. (04 Marks)

OR

- 10 a. Explain the following :
i) KIADB
ii) KSSIDC
iii) NSIC
iv) KSFC
v) DIC. (10 Marks)
b. Discuss the case study of Infosys – N.R. Narayana Murthy. (06 Marks)
c. List the advantages of Micro and Small enterprises. (04 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023 Computer Networks and Security

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the use of cookie files in web applications. (06 Marks)
- b. With a neat diagram, explain how SMTP can be used for transmitting mails from sender to receiver. (08 Marks)
- c. Discuss the working of Bit Torrent for file distribution. (06 Marks)

OR

- 2 a. Differentiate between persistent and non persistent connections in HTTP. (05 Marks)
- b. In brief explain the conditional GET operation. (05 Marks)
- c. Describe the DNS records and messages in detail. (10 Marks)

Module-2

- 3 a. In brief describe UDP segment structure and checksum computation. (06 Marks)
- b. With a neat diagram demonstrate the working of GO-BACK-N protocol. (08 Marks)
- c. Explain TCP flow control in detail. (06 Marks)

OR

- 4 a. With the help of a FSM, describe reliable data transfer in a Lossy channel with bit errors (rdt 3.0). (08 Marks)
- b. Explain the various fields of a TCP segment structure. (05 Marks)
- c. What are the approaches to congestion control? Explain in detail with example. (07 Marks)

Module-3

- 5 a. Explain inter autonomous system routing with Border Gateway protocol. (08 Marks)
- b. Explain various Broadcast Routing algorithms. (08 Marks)
- c. Write a note on IGMP protocol. (04 Marks)

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- 6 a. Write the link state algorithm and apply it to the following graph. Assume node 'u' as the source node. (10 Marks)

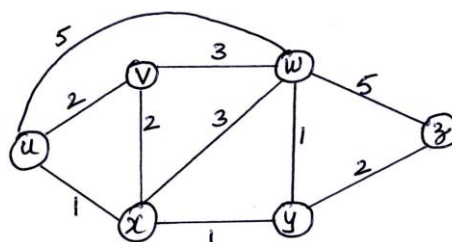


Fig.Q.6(a)

- b. Explain the architecture of a Router.

(10 Marks)

Module-4

- 7 a. What are the elements of network security? Discuss the threats to network security. (10 Marks)
- b. Explain RSA algorithm. Using RSA encrypt a message $m = 9$. Assume $p = 3$ and $q = 11$. Find the public key and private key, also show encryption and decryption. (10 Marks)

OR

- 8 a. Explain the working of DES algorithm. (08 Marks)
- b. Discuss the secure Hash Algorithm. (06 Marks)
- c. Write a note on firewalls. (06 Marks)

Module-5

- 9 a. Explain the types of multimedia network applications. (06 Marks)
- b. Briefly explain how DNS redirects a user request to a CDN server. (08 Marks)
- c. With a diagram, explain SIP call establishment. (06 Marks)

OR

- 10 a. What are the properties of video and audio? Explain in detail. (07 Marks)
- b. With a neat diagram, explain streaming stored video over HTTP. (07 Marks)
- c. Explain the Forward Error Correction (FEC) technique for loss anticipation in VoIP application. (06 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Database Management System

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe the characteristics of database approach. (08 Marks)
- b. List and explain the criteria for classification of DBMS. (08 Marks)
- c. Write an ER diagram to represent CAR entity type with 2 key attributes Registration and Vehicle ID. (04 Marks)

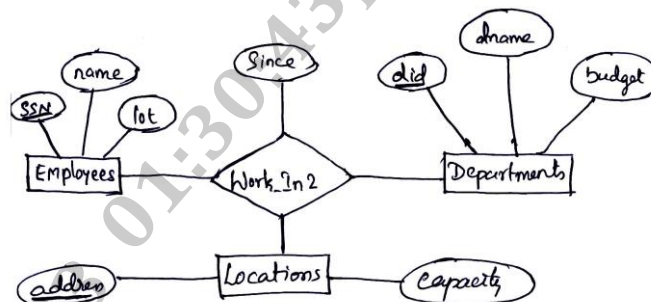
OR

- 2 a. Write an UML class diagram notation for company conceptual schema. (10 Marks)
- b. Define the following terms : i) Data Model ii) Schema iii) Instance
iv) Canned transaction v) Data Manipulation Language (DML). (10 Marks)

Module-2

- 3 a. Explain the concepts of specialization and Generalization, with the help of VEHICLE superclass. (08 Marks)
- b. Explain the different Relational Model constraints. (06 Marks)
- c. Create a table for the Works_In relationship shown in Fig. Q3(c). (06 Marks)

Fig. Q3(c)



OR

- 4 Considered the COMPANY DATABASE
 EMPLOYEE (Fname , Minit , Lname , Ssn , Bdate , Address , Sex , Salary , Super_Ssn , Dno).
 DEPARTMENT (Dname , Dnumber , Mgr_Ssn , Mgr_Start_data)
 DEPART_LOCATIONS (Dnumber , DLocation)
 PROJECT (Pname , Pnumber , PLocation , Dnum)
 WORKS_ON (ESsn , Pno , Hours)
 DEPENDENT (ESsn , Dependent_name , Sex , Bdate , Relationship).
 Specify the following queries in SQL on the database schema given above.
 - a. For every project located in 'Stafford' , list the project number, the controlling department number and the department managers last name , address and birth date. (06 Marks)
 - b. Retrieve the birth date and address of the employees whose name is 'John B, Smith'. (06 Marks)
 - c. Retrieve the name and address of all employees who work for the 'Research' department. (06 Marks)
 - d. Retrieve the salary of every employee. (02 Marks)

Module-3

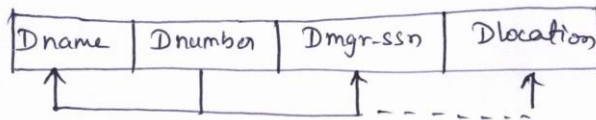
- 5 a. Discuss EXISTS and UNIQUE functions in SQL. Consider the COMPANY database given in Question number 04, write a query to list the name of the Manager who have atleast one dependent. (08 Marks)
- b. With a real World example, explain the following :
 i) JDBC ii) Correlated nested queries iii) Stored Procedures
 iv) Schema change statements in SQL. (12 Marks)

OR

- 6 a. Explain the usage of Aggregate function in SQL. Write an SQL query to find sum of the salaries of all employees, the maximum salary, the minimum salary and the average salary by renaming the columns in a single row table. (10 Marks)
- b. Create an HTML form to collect user id and password fields and it also has to have two buttons one for reset and another for login. (06 Marks)
- c. Write a short note on : i) JavaScript ii) CGI. (04 Marks)

Module-4

- 7 a. Describe the 3 main techniques to achieve first normal form for the relation by taking following examples schema. (04 Marks)



- b. Discuss the Informal guidelines to determine the quality of relations schema design with a suitable example. (10 Marks)
- c. Discuss the Insertion , Deletion and Modification anomalies. Illustrate, why are they considered bad, with an example. (06 Marks)

OR

- 8 a. What do you mean by Normalization? Explain 2NF and BCNF, with a suitable example. (06 Marks)
- b. Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies
 $F = \{\{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\}\}.$
 i) What is key of R? ii) Decompose R into 2NF and then 3NF relation. (06 Marks)
- c. Write an algorithm to find a minimal cover F for a set of functional dependencies E. (08 Marks)

Module-5

- 9 a. Discuss the ACID properties of database transaction. (08 Marks)
- b. Why concurrency control is needed? Demonstrate with an example. (12 Marks)

OR

- 10 a. Briefly explain 2 phase locking protocols. (05 Marks)
- b. Explain Transaction support in SQL. (05 Marks)
- c. Write a short note on :
 i) Single user and Multiuser system.
 ii) Transaction roll back and Cascading roll back.
 iii) Shadow paging.
 iv) Database backup and recovery from catastrophic failure.
 v) Deadlock prevention protocol. (10 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Automata Theory and Computability

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define DFSM. Design DFSM
 - i) To accept strings over $\{a, b\}$ such that each block of 5 (length five) consecutive symbols have atleast two a's.
 - ii) To accept $L = \{\omega(ab + ba) \mid \omega \in \{a, b\}^*\}$
 - iii) To accept $L = \{\omega bab \mid \omega \in \{a, b\}^*\}$
- b. Define distinguishable and indistinguishable states. Minimize the following DFSM.

δ	0	1
$\rightarrow A$	B	A
B	A	C
C	D	B
* D	D	A
E	D	F
F	G	E
G	F	G
H	G	D

(10 Marks)

OR

- 2 a. Convert the following NDFSM to DFSM. [Refer Fig.Q2(a)].



Fig.Q2(a)

(08 Marks)

- b. Explain the simulators for Finite State Machine.
- c. Design
 - (i) Mealy Machine that accepts the string that ends either with aa or bb and $\Sigma = \{a, b\}$
 - (ii) Moore Machine that produces 'A', 'B' and 'C' depending on inputs that end with '10', '11' and others respectively.

(06 Marks)

Module-2

- 3 a. Build regular expression from the following FSM. [Refer Fig.Q3(a)].

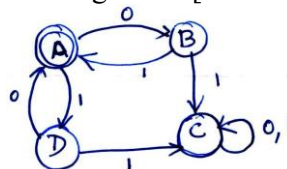


Fig.Q3(a)

(05 Marks)

- b. State and prove pumping Lemma theorem for regular languages. Show that $L = \{a^n b^n \mid n \geq 0\}$ is not Regular.
- c. Show that regular languages are closed under complement and intersection.

(10 Marks)

(05 Marks)

OR

- 4 a. Obtain Regular Expression for the following languages.
- (i) $L = \{ a^n b^m c^p \mid n \leq 4, m \geq 2, p \leq 2 \}$
 - (ii) $L = \{ \omega : |\omega| \bmod 3 = 0 \text{ \& } \omega \in \{a, b\}^* \}$
 - (iii) $L = \{ a^n b^m \mid m + n \text{ is even} \}$ (08 Marks)
- b. Prove Kleen's theorem – Any language that can be defined with a regular expression can be accepted by some FSM and so is regular. (08 Marks)
- c. Obtain NDFSM for the following regular expression $(a + b)^* abb$. (04 Marks)

Module-3

- 5 a. Design a PDA for the language
 $L = \{ \omega \omega^R \mid \omega \in (a, b)^* \text{ where } \omega^R \text{ is reverse of } \omega \}$
 and show the moves made by PDA for the string "aabcbaa" and "abacbbba". (10 Marks)
- b. Define Leftmost derivation, Rightmost derivation and Parse tree. Consider the grammar.
- $$S \rightarrow AbB \quad A \rightarrow aA \mid \epsilon$$
- $$B \rightarrow aB \mid bB \mid \epsilon \quad D \rightarrow a \mid \epsilon$$
- Obtain LMD, RMD and parse tree for the string "aaabab". (10 Marks)

OR

- 6 a. Define CFG and design a CFG for the following language.
- (i) $L = \{ 0^m 1^m 2^n \mid m \geq 1 \text{ and } n \geq 0 \}$
 - (ii) $L = \{ \omega \omega^R \mid \omega \in (a, b)^* \}$
 - (iii) $L = \{ a^n b^m c^k \mid n+2m = k \text{ for } m \geq 0 \text{ and } n \geq 0 \}$ (10 Marks)
- b. Define CNF. Convert the following CFG into CNF.
- $$S \rightarrow ASB \mid \epsilon \quad A \rightarrow aAS \mid a \quad B \rightarrow SbS \mid A \mid bb$$
- (10 Marks)

Module-4

- 7 a. Define TM and design a turing machine for $L = \{ \omega \mid \omega \in (0+1)^* \text{ containing the substring } 001 \}$
 Write transition diagram and show the moves made by the Turing machine for input string 10010. (14 Marks)
- b. Define and explain DTM and NDTM. (06 Marks)

OR

- 8 a. With a neat diagram explain the working of Multitape Turing Machine. (08 Marks)
- b. Design a Turing machine to accept $L = \{ 0^n 1^n \mid n \geq 1 \}$. Show the moves made for the string 0011 and 00111. (12 Marks)

Module-5

- 9 Write short notes on :
- a. Linear Bound Automata (06 Marks)
 - b. Church Turing Thesis (07 Marks)
 - c. Non-Deterministic Turing Machine (07 Marks)

OR

- 10 a. Explain Halting Problem and Post Correspondence problem in Turing Machine. (10 Marks)
- b. Discuss the following :
- i) Decidable and Undecidable Language (05 Marks)
 - ii) Quantum Computers (05 Marks)

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18CS55

Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Application Development using Python

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1
 - a. List the salient features of python programming language. (04 Marks)
 - b. Explain the math operators in python from highest to lowest precedence with an example for each. Show steps to evaluate $(5 - 1) * (7 + 1) (3 - 1)$ in python. (08 Marks)
 - c. Write a program to find the sum of all odd and even numbers of n elements. Here skip the numbers which are divisible by 3. (08 Marks)

OR

- 2
 - a. Give one example for each of the operation below in python language:
 - i) String concatenation and replication.
 - ii) Read input and display. (10 Marks)
 - b. Explain:
 - i) Def statements with parameters
 - ii) Return values and return statements with an example. (10 Marks)

Module-2

- 3
 - a. Discuss the different ways of traversing a list. Explain each with an example. (10 Marks)
 - b. Write a python program that allows a player to guess a secret number within 6 chances. The code that lets the player enter a guess and checks that guess is right or not by printing appropriate message. List of numbers are taken as an input from the user. (07 Marks)
 - c. Write a program to demonstrate the use of pretty function. (03 Marks)

OR

- 4
 - a. Compare list and dictionary data structures with respect to python language. (04 Marks)
 - b. Write a program in python that counts. The number of occurrences of each letter in a string. Display the results in column fashion. (08 Marks)
 - c. Write the string method syntax in python to perform below operations.
 - i) Removing white space characters from beginning, end or both sides of a string.
 - ii) To right-justify, left-justify and center a string. (08 Marks)

Module-3

- 5
 - a. List out the different character classes. Give representation, regular expression symbols, example and meanings for each character class. (10 Marks)
 - b. Describe the following with suitable python code snippet:
 - i) Greedy and Non Greedy pattern matching.
 - ii) Findall() method of RegeX object. (10 Marks)

OR

- 6 a. Write a python program to rename the filename contains American style dates (MM – DD – YYYY) to European style dates (DD – MM – YYYY) in the working directory. (10 Marks)
- b. What are Assertions in python? Explain with an example. (05 Marks)
- c. Explain the file reading and writing process with suitable python program. (05 Marks)

Module-4

- 7 a. How objects are mutable by nature justify with an example? (04 Marks)
- b. Discuss the methods `isinstance()` and `hasattr()` with suitable example for each. (06 Marks)
- c. What is operator over loading? Write a program to add two point objects by overloading + operator. Also, overload – str – () to display point as an ordered pair. (10 Marks)

OR

- 8 a. Define Inheritance. Explain with an example. (06 Marks)
- b. Briefly discuss the importance of - `__init__()` and - `__str__()` methods in python. (04 Marks)
- c. Demonstrate the polymorphism to generate histogram to count the number of times each letter appears in word and in sentence. (10 Marks)

Module-5

- 9 a. Analyze the steps involved in downloading and saving web page on to local system along with program. (06 Marks)
- b. List any 4 CSS selectors of bs4 module. Using Beautiful soup passel, retrieve all of the paragraph tags in the web page www.amazon.com. (06 Marks)
- c. How selenium module is useful to deal with web pages. What methods do it uses to simulate mouse clicks and keyboard keys? (08 Marks)

OR

- 10 a. Write a program to read the census data from the excel spreadsheet, count the number of census tracts in each country, count the total population of each country and prints the results.

1	Census tract	State	County	POP2010
9841	06075010500	CA	SanFrancisco	2685
9842	06075010600	CA	SanFrancisco	3894
9843	06075010700	CA	SanFrancisco	5592

- (08 Marks)
- b. How to zip the files and folders. Demonstrate with one example. (04 Marks)
- c. Write a script that will go through every PDF in a folder and encrypt the PDFs using a password provided on the command line. Save each encrypted PDF with an-encrypted pdf suffix added to the original filename. (08 Marks)

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18CS56

Fifth Semester B.E. Degree Examination, Jan./Feb. 2023 UNIX Programming

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Compare internal commands and external commands, files and processes. (06 Marks)
b. Explain all the features of UNIX operating system. (09 Marks)
c. Write the output for the following commands :
i) cal 10 2021
ii) date +%D%T"
iii) type echo
iv) passwd
v) who. (05 Marks)

OR

- 2 a. Explain the different categories of files with examples. (06 Marks)
b. Describe the parent child relationship in UNIX file system and differentiate absolute pathnames with relative path names. (06 Marks)
c. Write the description for the following commands.
i) mkdir college college/ISE college/CSE
ii) mv f₁.C f₂.C f₃.C cprogs
iii) if my pwd is /home/ravi/progs then Cd ../.
iv) ls -l | wc -l
v) cp f₁ f₂ f₃ files
vi) rm -i chap1
vii) cat >> test.txt
viii) rmdir college/ISE (08 Marks)

Module-2

- 3 a. Explain all the options of ls commands with examples. (06 Marks)
b. Consider a file test.txt with default permissions as -rw-r--r--, grant execute permission to owner, write and execute permission to group members and execute permission to others using both relative and absolute approaches. (04 Marks)
c. Write the output for the following commands.
i) cp ??? progs
ii) rm 'chap*'
iii) mv *.[!C][!P][!P] progs
iv) cat *.txt | wc -C
v) cp chap\[0-1\]. (05 Marks)
d. Explain the grep command with all its options. (05 Marks)

OR

- 4 a. Write a program to read pattern and filename from the user and search the pattern in the given file. (05 Marks)
- b. Write the output for the following commands.
- i) `grep "Anil" std.lst || echo "pattern not found"`
 - ii) `test $x -gt $y`
 - iii) `[-Z $stg]`
 - iv) `[-r $file]`
 - v) `[!-n $stg]` (05 Marks)
- c. Explain all the looping statements with syntax. (06 Marks)
- d. Write a shell script to read multiple patterns from the command line and search these patterns in the given file which is also read from command line by using shift command. [Ex. Command line arguments as below `#>script.sh pat1 pat2 pat3 pat4 pat5`]. (04 Marks)

Module-3

- 5 a. Explain the General File API's `open()`, `read()`, `write()`, `lseek()` with their prototype. (10 Marks)
- b. Describe the memory layout of a C program with a diagram and explain memory allocation API's with their prototypes. (10 Marks)

OR

- 6 a. Explain `setjmp` and `longjmp`, `getrlimit` and `setrlimit` function with examples. (10 Marks)
- b. Describe how the process is created by using `fork()` and `vfork()`. List out the properties inherited from the parent when the child process is created? (10 Marks)

Module-4

- 7 a. Explain the implementation of system function using `fork()`, `exec()`, `wait()` API's. (10 Marks)
- b. Define pipes, write a program to send data from parent to child using pipe API and also list its limitations. (10 Marks)

OR

- 8 a. Define semaphores and explain how the IPC is implemented using various semaphore API's. (10 Marks)
- b. Explain the implementation of shared memory IPC mechanism with all its API's and their prototypes. (10 Marks)

Module-5

- 9 a. Define signal and list the actions taken by a process when the signal is raised. Explain the signal API's `signal()`, `sigset()`, `sigaction()`. (10 Marks)
- b. Explain how `kill` API is used for sending a signal to a process and explain the implementation of sleep API using alarm API. (10 Marks)

OR

- 10 a. Define the Daemon process. Explain all the coding rules to be followed while coding a daemon process (10 Marks)
- b. Write a note on interval timer. (05 Marks)
- c. Explain the BSD syslog facility for handling Daemons error messages. (05 Marks)

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Question Paper Version : A

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Which of the following is a possible producer in an ecosystem?
a) Animal b) Plants c) Human beings d) Fish
 2. The largest reservoir of nitrogen in our planet is
a) Oceans b) Biosphere c) Atmosphere d) Rivers
 3. India has the world's largest share of which of the following
a) Manganese b) Mica c) Copper d) Diamond
 4. Identify the non renewable source of energy from the following:
a) Coal b) Fuel cells c) Wind power d) Wave power
 5. Which of the following terminologies is not associated with the vertical structure of forest?
a) Canopy b) Understory c) Forest floor d) First floor
 6. Which of the following is cause of loss of biodiversity?
a) Habitat degradation b) Invasion of non-native species
c) Pollution d) All of these
 7. Air pollution from automobiles can be controlled by fitting
a) Electrostatic precipitator b) Cyclone separator
c) Wet collector d) Catalytic converter
 8. When the solid waste consists of large amount of organic matter and if the moisture content is high, which of the following methods of treatment will be ideal?
a) Composting b) Palletizing
c) Incineration d) Recycling

9. Chernobyl Nuclear Disaster occurred in the year
a) 1984 b) 1985 c) 1986 d) 1987
10. The primary cause of acid rain around the world is
a) Carbon dioxide b) Sulphur dioxide
c) Carbon monoxide d) Ozone
11. World Environmental day is held every year on
a) June 5th b) October 2nd c) April 22nd d) November 1st
12. Ozone layer thickness is measured in _____.
a) mm b) cm c) Dobson unit d) Db
13. First of the major environmental protection acts to be promulgated in India was
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a) USA b) Russia
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94. Which among the following has highest percentage of calorific value?
a) Anthracite b) Peat
c) Lignite d) Bituminous coal
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a) Leaf b) Stem
c) Roots d) Flower
96. The two major components of ecosystem are
a) Adiabatic and isotropic b) Ecologic and climatologic
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97. Geothermal energy is a
a) Heat energy b) Wind energy
c) Current energy d) Solar energy
98. The average life expectancy around the world is currently.
a) Decreasing b) Increasing
c) Stabilizing d) Not changing
99. The universal declaration of Human Rights was proclaimed by the UN in the year.
a) 1946 b) 1947
c) 1948 d) 1949
100. The objective of Integrated Child Development Service (ICDS) are
a) Immunization b) Health check up and referral services
c) Pre-school non formal education d) All of these

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Question Paper Version : B

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

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40. Which of the following element make e-waste hazardous in nature?
a) Land
b) Glass
c) Plastic
d) Iron
41. What is the hazardous pollutant released from batteries?
a) Arsenic
b) Barium
c) Cobalt
d) Cadmium
42. What is the term used for reuse of sewage sludge?
a) Compost
b) Solids
c) Biosolids
d) Sludge
43. Reduction in brightness of the famous Taj Mahal is due to
a) Global warming
b) Air pollution
c) Ozone depletion
d) Afforestation
44. E.I.A. can be expanded as
a) Environment and Industrial Act
b) Environment of Impact Activities
c) Environmental Impact Assessment
d) Environmentally Important Activity
45. Organic Farming is
a) Farming without using pesticides and chemical fertilizer
b) Enhances biodiversity
c) Promotes soil biological activity
d) All of these
46. Bio-remediation means the removal of contaminants from
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- Version - B - 5 of 8

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- 1 Nuclear Disaster occurred in Japan in
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Question Paper Version : C

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

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30. Which of the following is a possible producer in an ecosystem?
a) Animal b) Plants c) Human beings d) Fish
31. The largest reservoir of nitrogen in our planet is
a) Oceans b) Biosphere c) Atmosphere d) Rivers
32. India has the world's largest share of which of the following
a) Manganese b) Mica c) Copper d) Diamond
33. Identify the non renewable source of energy from the following:
a) Coal b) Fuel cells c) Wind power d) Wave power
34. Which of the following terminologies is not associated with the vertical structure of forest?
a) Canopy b) Understory c) Forest floor d) First floor
35. Which of the following is cause of loss of biodiversity?
a) Habitat degradation b) Invasion of non-native species
c) Pollution d) All of these

36. Air pollution from automobiles can be controlled by fitting
a) Electrostatic precipitator b) Cyclone separator
c) Wet collector d) Catalytic converter
37. When the solid waste consists of large amount of organic matter and if the moisture content is high, which of the following methods of treatment will be ideal?
a) Composting b) Palletizing
c) Incineration d) Recycling
38. Chernobyl Nuclear Disaster occurred in the year
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c) Rio-de-Janeiro d) Johannesburg
63. Which among the following has highest percentage of calorific value?
a) Anthracite b) Peat
c) Lignite d) Bituminous coal
64. Nitrogen fixing bacteria exists in
a) Leaf b) Stem
c) Roots d) Flower

65. The two major components of ecosystem are
a) Adiabatic and isotropic
b) Ecologic and climatologic
c) Cyclic and biologic
d) Abiotic and biotic
66. Geothermal energy is a
a) Heat energy
b) Wind energy
c) Current energy
d) Solar energy
67. The average life expectancy around the world is currently.
a) Decreasing
b) Increasing
c) Stabilizing
d) Not changing
68. The universal declaration of Human Rights was proclaimed by the UN in the year.
a) 1946
b) 1947
c) 1948
d) 1949
69. The objective of Integrated Child Development Service (ICDS) are
a) Immunization
b) Health check up and referral services
c) Pre-school non formal education
d) All of these
70. South Africa is leading exporter of which mineral?
a) Copper
b) Diamond
c) Silver
d) Gold
71. The word 'sustainable development' came into existence in the year.
a) 1992
b) 1978
c) 1980
d) 1987
72. The other word of landscaping is
a) Reduction
b) Restoration
c) Removing topsoil
d) Restore
73. Cloud seeding with silver iodide is based on the
a) Bergeron process
b) Collision-coalescence process
c) Both a and b
d) None of these
74. Environmental pollution is due to
a) Rapid urbanization
b) Deforestation
c) Afforestation
d) a and b
75. The liquid waste from bathroom and kitchen is called
a) Sullage
b) Domestic sewage
c) Storm water
d) Runoff
76. BOD means
a) Biochemical Oxygen Demand
b) Chemical oxygen demand
c) Biophysical Oxygen Demand
d) All of these
77. Which of the following source is surface water?
a) Springs
b) Streams
c) Deep wells
d) All of these
78. Which of the following is biodegradable?
a) Plastics
b) Domestic sewage
c) Detergents
d) a and c

79. Blaring sounds known to cause
 a) Mental distress
 b) High cholesterol
 c) Neurological problems
 d) All of these
80. In geosynchronous orbit altitude of the satellite is about
 a) 36,000 kms
 b) 10,000 kms
 c) 50,000 kms
 d) None of these
81. The Air (Prevention and Control of Pollution) Act was enacted in the year.
 a) 1987
 b) 1981
 c) 1991
 d) 1988
82. Kudremukh Iron ore mine, Karnataka was closed due to
 a) River pollution and threat to biodiversity
 b) Land encroachment
 c) Radioactive hazards
 d) Serious health hazard
83. On the eve of Gandhi Jayanthi which andolan was launched by our Honorable Prime Minister
 a) Swadeshi
 b) Sarvashikshana Abhiyana
 c) Suvarnagrama
 d) Swach Bharath
84. An international agreement signed in the year 1987, to protect stratospheric ozone is known as
 a) Montreal protocol
 b) Kyoto protocol
 c) Earth summit
 d) None of these
85. The explosion of First Atomic Bomb was done in Hiroshima and Nagasaki in
 a) 1946
 b) 1986
 c) 1945
 d) 1947
86. A dangerous pesticide which has been reported to cause physical deformities to people of Kerala and Karnataka states
 a) Endosulfan
 b) Fluorides
 c) DDT
 d) Dioxygene
87. Visible portion of EMR ranges between
 a) 0.4 - 0.76 μm
 b) 10.5 - 12.5 μm
 c) 8.0 - 14.0 μm
 d) None of these
88. Data representation in Raster data is by
 a) pixel
 b) points, lines and polygon
 c) latitude and longitude
 d) none of these
89. In water treatment, alum is used for
 a) softening
 b) coagulation
 c) filtration
 d) disinfection
90. Which of the following element make e-waste hazardous in nature?
 a) Land
 b) Glass
 c) Plastic
 d) Iron
91. What is the hazardous pollutant released from batteries?
 a) Arsenic
 b) Barium
 c) Cobalt
 d) Cadmium
92. What is the term used for reuse of sewage sludge?
 a) Compost
 b) Solids
 c) Biosolids
 d) Sludge

93. Reduction in brightness of the famous Taj Mahal is due to
a) Global warming
b) Air pollution
c) Ozone depletion
d) Afforestation
94. E.I.A. can be expanded as
a) Environment and Industrial Act
b) Environment of Impact Activities
c) Environmental Impact Assessment
d) Environmentally Important Activity
95. Organic Farming is
a) Farming without using pesticides and chemical fertilizer
b) Enhances biodiversity
c) Promotes soil biological activity
d) All of these
96. Bio-remediation means the removal of contaminants from
a) Soil
b) Wastewater
c) Groundwater
d) Both soil and ground water
97. Plants use _____ gas for photosynthesis.
a) Oxygen
b) Methane
c) Nitrogen
d) Carbon dioxide
98. What is the maximum allowable concentration of fluorides in drinking water?
a) 1.0 mg/l
b) 1.25 mg/l
c) 1.50 mg/l
d) 1.75 mg/l
99. Forest rich area in Karnataka is found in _____.
a) Western Ghats
b) Bandipur
c) Nagarhole
d) Mangalore

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Question Paper Version : D

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies

(COMMON TO ALL BRANCHES)

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. World Environmental day is held every year on
a) June 5th b) October 2nd c) April 22nd d) November 1st
 2. Ozone layer thickness is measured in _____.
a) mm b) cm c) Dobson unit d) Db
 3. First of the major environmental protection acts to be promulgated in India was
a) The Water Act b) The Air Act
c) The Environment Act d) Noise Pollution Rules
 4. Blue baby syndrome is caused due to _____.
a) Manganese b) Ozone c) Silver d) Nitrate
 5. World Earth's day is annually celebrated on
a) April 22nd b) June 5th c) January 1st d) May 1st
 6. The most important fuel used by nuclear power plant is
a) U-235 b) U-238 c) U-245 d) U-248
 7. Which of the following is a biotic component of ecosystem?
a) Fungi b) Solar light
c) Temperature d) Humidity
 8. Abiotic component includes
a) Soil b) Temperature
c) Water d) All of these

9. The word "Environment" is derived from
 - a) Greek
 - b) French
 - c) Spanish
 - d) English
10. Which of the following is absorbed by green plants from the atmosphere?
 - a) Carbon dioxide
 - b) Water
 - c) Nutrients
 - d) All of these
11. South Africa is leading exporter of which mineral?
 - a) Copper
 - b) Diamond
 - c) Silver
 - d) Gold
12. The word 'sustainable development' came into existence in the year.
 - a) 1992
 - b) 1978
 - c) 1980
 - d) 1987
13. The other word of landscaping is
 - a) Reduction
 - b) Restoration
 - c) Removing topsoil
 - d) Restore
14. Cloud seeding with silver iodide is based on the
 - a) Bergeron process
 - b) Collision-coalescence process
 - c) Both a and b
 - d) None of these
15. Environmental pollution is due to
 - a) Rapid urbanization
 - b) Deforestation
 - c) Afforestation
 - d) a and b
16. The liquid waste from bathroom and kitchen is called
 - a) Sullage
 - b) Domestic sewage
 - c) Storm water
 - d) Runoff
17. BOD means
 - a) Biochemical Oxygen Demand
 - b) Chemical oxygen demand
 - c) Biophysical Oxygen Demand
 - d) All of these
18. Which of the following source is surface water?
 - a) Springs
 - b) Streams
 - c) Deep wells
 - d) All of these
19. Which of the following is biodegradable?
 - a) Plastics
 - b) Domestic sewage
 - c) Detergents
 - d) a and c
20. Blaring sounds known to cause
 - a) Mental distress
 - b) High cholesterol
 - c) Neurological problems
 - d) All of these
21. Which of the following element make e-waste hazardous in nature?
 - a) Land
 - b) Glass
 - c) Plastic
 - d) Iron
22. What is the hazardous pollutant released from batteries?
 - a) Arsenic
 - b) Barium
 - c) Cobalt
 - d) Cadmium

- _____ gas for photosynthesis.
- a) Methane
b) Carbon dioxide
- c) The maximum allowable concentration of fluorides in drinking water is _____ mg/l.
- a) 1.25 mg/l
b) 1.75 mg/l
- c) The area in Karnataka is found in _____.
- a) Bandipur
b) Mangalore
- c) The synchronous orbit altitude of the satellite is about _____ kms.
- a) 10,000 kms
b) None of these
- c) The Prevention and Control of Pollution) Act was enacted in _____.
- a) 1981
b) 1991
- c) The Iron ore mine, Karnataka was closed due to _____.

- a) 1982
 b) 1982
 c) 1973
 d) 1973
- the following is not a part of the hydrological cycle?
 a) Evaporation
 b) Infiltration
 c) Precipitation
 d) Perspiration
- International Earth Summit was held at
 a) Rio de Janeiro
 b) Russia
 c) New York
 d) Johannesburg
- Among the following has highest percentage of calorific value
 a) Lignite
 b) Peat
 c) Anthracite
 d) Bituminous coal
- Nitrogen fixing bacteria exists in
 a) Root
 b) Stem
 c) Leaf
 d) Flower
- Major components of ecosystem are
 a) Biotic and isotropic
 b) Ecologic and c
 c) Biotic and biotic
 d) Abiotic and b

- al declaration of Human Rights was proclaimed by the
- b) 1947
- d) 1949
- e of Integrated Child Development Service (ICDS) are
- b) Health check up and
- d) All of these
- e following is a possible producer in an ecosystem?
- b) Plants
- c) Human beings
- reservoir of nitrogen in our planet is
- b) Biosphere
- c) Atmosphere
- e world's largest share of which of the following
- b) Mica
- c) Copper
- non renewable source of energy from the following:
- b) Fuel cells
- c) Wind power
- e following terminologies is not associated with the
- b) Understory
- c) Forest floor
- e following is cause of class of biodiversity?
- b) Invasion of non-na
- d) All of these
- n from automobiles can be controlled by fitting
- b) Cyclone separator
- d) Catalytic converter
- olid waste consists of large amount of organic matter
- b) Palletizing
- d) Recycling
- Nuclear Disaster occurred in the year
- b) 1985
- c) 1986
- cause of acid rain around the world is
- b) Sulphur dioxide
- d) Ozone

63. Which among the following is a climatic factor?
 a) pressure b) humidity
 c) temperature d) all of these
64. Biodiversity can be broadly classified into how many types?
 a) 2 b) 5 c) 3 d) 4
65. Hot spot areas have
 a) Low density of biodiversity b) Only endangered plants
 c) High density of hot springs d) High density of biodiversity
66. About _____% of the earth's surface is covered by water.
 a) 53% b) 19% c) 71% d) 90%
67. Deforestation means
 a) preservation of forests b) destruction of forests
 c) monocrop cultivation d) agriculture
68. When did National Disaster Management Authority formed?
 a) 2000 b) 2005 c) 2010 d) 2015
69. Disaster is an event arising out of
 a) result of hazard event b) causes of hazard event
 c) causes of disaster event d) all of these
70. The scientific study of earthquake is called
 a) seismograph b) seismology
 c) both a and b d) none of these
71. "Minamata Disease" is caused due to
 a) Lead b) Arsenic c) Mercury d) Cadmium
72. Alternative eco-friendly fuel for automobiles is
 a) Petrol b) Diesel c) CNG d) Kerosene
73. Population explosion will cause
 a) Biodiversity b) Stress on ecosystem
 c) More employment d) None of these
74. Which of the following is having high population density?
 a) India b) China c) USA d) Western Europe
75. Demography is the study of
 a) Animals behaviour b) Population growth
 c) River d) None of these
76. Forest are called as _____.
 a) Air purifier b) Earth's lungs c) Oxygen reservoir d) CO₂ absorbers
77. Which of the following is the facility that the urban people enjoy?
 a) Better quality of air b) Better communication access
 c) Large land at cheap rates d) None of these

78. Which of the following is an air pollutant?
 a) Carbon dioxide
 b) Oxygen
 c) Nitrogen
 d) Particulate matter
79. Cyto toxic and expired drugs are disposed of by
 a) damping
 b) autoclave
 c) incineration
 d) chemical disinfection
80. The colour code of plastic bag for disposing of microbial laboratory culture waste.
 a) Black
 b) Red
 c) Blue
 d) White
81. Among the fresh water available in the earth the percentage of surface water is about
 a) 50%
 b) 10%
 c) 5%
 d) less than 1%
82. Hepatitis is caused by
 a) Protozoa
 b) Virus
 c) Bacteria
 d) Fungus
83. In India groundwater resources are rich in
 a) Plains of river Kaveri and Krishna
 b) The Deccan plateau
 c) The Gangetic plains
 d) The plains of Netravati and Kapila
84. The required iron content in drinking water as specified by BIS is
 a) 300 mg/l
 b) 30 mg/l
 c) 3 mg/l
 d) 0.30 mg/l
85. Molasses from sugar industry is used to generate
 a) Biodiesel
 b) Hydrogen
 c) Bioethanol
 d) Biomethanol
86. Wind Farms are located in
 a) River basin
 b) Plain area
 c) Hilly area
 d) Valley area
87. Biomass consists of
 a) Lignin
 b) Hemi cellulose
 c) Cellulose
 d) All of these
88. Natural gas contains
 a) Carbon dioxide
 b) Hydrogen
 c) Methane
 d) Nitrogen
89. Anti tobacco day is mentioned on
 a) 31st May
 b) 30st June
 c) 31st July
 d) 31st August
90. Population explosion will cause
 a) Socio-Economic Problems
 b) Food Scarcity
 c) Energy crises
 d) All of these
91. GIS stands for
 a) Geostationary Interact Sector
 b) Geographical Information System
 c) Geotechnical Information Society
 d) Geothermal Investigation Site

92. LPG is a mixture of
a) N_2 and H_2S b) CO_2 and N_2
c) Propane and butanes d) Methane and CO_2
93. The Tiger Conservation Project was started in
a) 1973 b) 1975 c) 1981 d) 2000
94. The leader of "Chipko Movement" is
a) Sunderlal Bahuguna b) Medha Patkar
c) Vandana Shiva d) Mahatma Gandhi
95. Which of the following is the source of Fly-ash?
a) Vehicular exhaust b) Sewage
c) Thermal power plant d) All of these
96. The permissible range of pH for drinking water as per the Indian Standard is
a) 6 to 9 b) 6.5 to 8.5
c) 6 to 8.5 d) 6.5 to 7.5
97. Water logging is a phenomena in which
a) Water patterns are rotated
b) Soil root zone becomes saturated due to over irrigation
c) Erosion of soil
d) Soil degradation
98. Carbon content is higher in
a) Living matter b) Soil
c) Water d) Atmosphere
99. Springs means
a) Surface water b) Atmosphere water
c) Both (a) and (b) d) Ground water
100. Bio-remediation means deliberately introducing micro organisms to break pollutants.
a) Soil b) Waste water
c) Ground water d) Both soil and groundwater

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CBCS SCHEME

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18CS71

Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

Artificial Intelligence and Machine Learning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. The water jug problem states : You are provided with two jugs, one with 4 gallons of capacity and the other one with 3 gallons of capacity. Neither have any measuring markers on it. How can we get exactly two gallons of water in 4 gallon jug?
 - (i) Write down the production rules for the above problem.
 - (ii) Write any one solution for the above problem. (08 Marks)
- b. Explain Steepest Ascent Hill Climbing technique with an algorithm. Comment on its drawbacks and how to overcome these drawbacks. (12 Marks)

OR

- 2 a. Explain problem reduction with respect to AND-OR graph with suitable example. (07 Marks)
- b. Write AO* algorithm. (07 Marks)
- c. Discuss about constraint satisfaction and solve the below crypt arithmetic problem.

$$\text{CROSS} + \text{ROADS} = \text{DANGER}$$
(06 Marks)

Module-2

- 3 Consider the following sentences :
 1. John likes all kinds of food
 2. Apples are food
 3. Chicken is food
 4. Anything anyone eats and isn't killed is food
 5. Bill eats peanuts and is still alive
 6. Sue eats everything Bill eats.
 - (i) Translate these sentences into formulas in predicate logic. (05 Marks)
 - (ii) Prove that John likes peanuts using backward chaining. (05 Marks)
 - (iii) Convert the formulas of (i) into clause form. (05 Marks)
 - (iv) Prove John likes peanuts using resolution. (05 Marks)

OR

- 4 a. Distinguish forward and backward reasoning with an example. (04 Marks)
- b. Find maximally specific hypothesis for the training instances given below. Also write Find-S algorithm. The concept of this particular problem will be on what days does a person lines to go on walk.

Time	Weather	Temperature	Company	Humidity	Wind	Goes
Morning	Sunny	Warm	Yes	Mild	Strong	Yes
Evening	Rainy	Cold	No	Mild	Normal	No
Morning	Sunny	Moderate	Yes	Normal	Normal	Yes
Evening	Sunny	Cold	Yes	High	Strong	Yes

(08 Marks)

- c. Define version space. Discuss the limitations of finds algorithm over candidate elimination algorithm. (08 Marks)

Module-3

- 5 a. Explain the concept of decision tree learning. Write about attribute selection measure used to build the decision tree using ID3 algorithm. (07 Marks)
- b. How a single perceptron can be used to represent the Boolean functions such as AND, OR. (06 Marks)
- c. Write Gradient Descent algorithm to train a linear unit along with the derivation. (07 Marks)

OR

- 6 a. What do you mean by Gain and entropy? How it is used to build the decision tree. (08 Marks)
- b. Explain back propagation algorithm. Why is it not likely to be trapped in local minima? (08 Marks)
- c. Discuss the perceptron training rule and delta rule that solves the learning problem of perceptron. (04 Marks)

Module-4

- 7 a. Explain Naïve Bayes classifier. (08 Marks)
- b. Explain Bayesian Belief network and conditional independence with example. (08 Marks)
- c. Let us assume dangerous fires are rare (1%) but smoke is fairly common (10%) due to barbecues, and 90% of dangerous fires make smokes. Find the probability of dangerous fire when there is smoke. (04 Marks)

OR

- 8 a. Discuss minimum description length principle in brief. (08 Marks)
- b. Explain brute force MAP learning algorithm. (08 Marks)
- c. Explain EM algorithm. (04 Marks)

Module-5

- 9 a. Explain k-Nearest neighbor learning algorithm. (08 Marks)
- b. Explain Locally weighted regression. (08 Marks)
- c. What is reinforcement learning? (04 Marks)

OR

- 10 a. Distinguish Eager learning vs Lazy learning algorithms. (04 Marks)
- b. Write short notes on Q-learning. (08 Marks)
- c. Discuss about Radial basis function in detail. (08 Marks)

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Seventh Semester B.E. Degree Examination, June/July 2023 Artificial Intelligence and Machine Learning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1. a. What is Artificial Intelligence? List the applications. (05 Marks)
- b. Solve the 4-gallon water jug problem, writing appropriate production rules. (07 Marks)
- c. Apply A* algorithm and find the most cost effective path from start state A to final state J. (08 Marks)

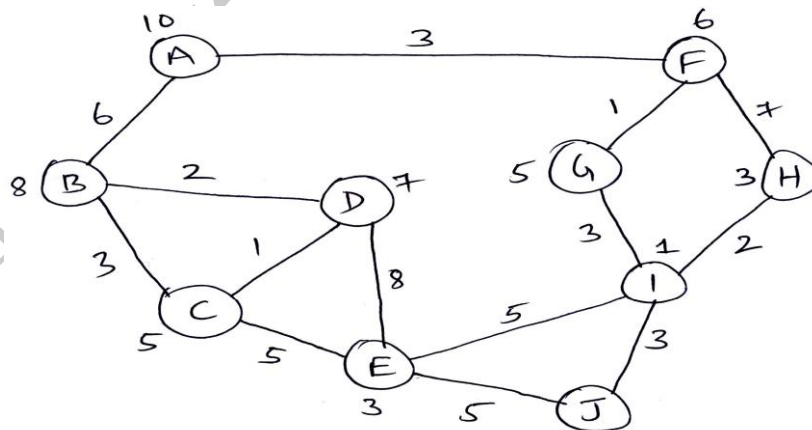


Fig. Q1 (c)

(08 Marks)

OR

2. a. What is Heuristic search? Explain the characteristics of heuristic search. (07 Marks)
- b. Provide the solution for water Jug problem by stating the general rules. (07 Marks)
- c. Write the Hill climbing search algorithm of AI. (06 Marks)

Module-2

3. a. Discuss the framework of knowledge representation along with any two knowledge representation scheme. (10 Marks)
- b. Explain in brief forward reasoning and backward reasoning. (10 Marks)

OR

4. a. Define concept learning. Explain the task of concept learning. (05 Marks)
- b. Write Find-S algorithm. (05 Marks)
- c. Write candidate elimination algorithm. Apply the algorithm for the below dataset to obtain the final version space.

Ex	Citations	Size	Inlibrary	Price	Editions	Buy
1	Some	Small	No	Affordable	One	No
2	Many	Big	No	Expensive	Many	Yes
3	Many	Medium	No	Expensive	Few	Yes
4	Many	Small	No	Affordable	Many	Yes

(10 Marks)

Module-3

- 5 a. Explain the concepts of entropy and information gain in decision tree. (05 Marks)
 b. Apply ID3 algorithm for constructing decision tree for the following training examples. (10 Marks)

Instances	a ₁	a ₂	a ₃	Classification
1	True	Hot	High	No
2	True	Hot	High	No
3	False	Hot	High	Yes
4	False	Cool	Normal	Yes
5	False	Cool	Normal	Yes
6	True	Cool	High	No
7	True	Hot	High	No
8	True	Hot	Normal	Yes
9	False	Cool	Normal	Yes
10	False	Cool	High	Yes

- c. List and explain the appropriate problems for decision tree learning. (05 Marks)

OR

- 6 a. Apply perceptron rule for implement XOR gate by considering the following and compute the final weights.

Inputs : X_1, X_2

Output : y

Initial weights : $W_{11} = W_{21} = 1$

$W_{12} = W_{22} = 1$

$V_1 = V_2 = 1$

Threshold = 1 and learning rate = 1.5

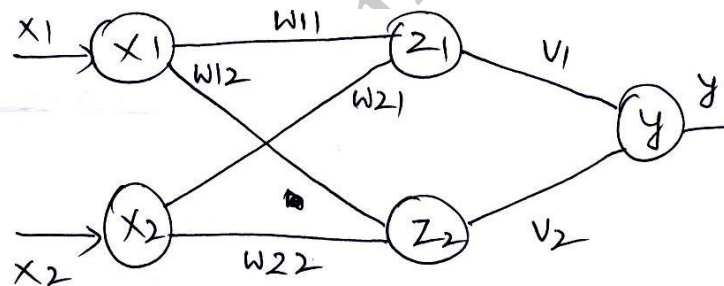


Fig. Q6 (a)

- b. Write Back propagation algorithm. (06 Marks)
 c. Discuss the perceptron training rule and delta rule that solves the learning problem of perceptron. (07 Marks)

Module-4

- 7 a. Explain Bayes theorem and MAP hypothesis with equations. (06 Marks)
 b. Outline Brute force MAP learning algorithm. (08 Marks)
 c. In Orange country, 51% of the adults are males (other 49% are ofcourse females). One adult is randomly selected for a survey involving credit card usage.
 (i) Find the prior probability that the selected person is male.
 (ii) It is later learned that the selected survey subject was Smoking a Cigar. Also, 9.5% of males smoke Cigar, whereas 1.7% of females smoke Cigars. Use this additional information to find the probability that the selected subject is a male. (06 Marks)

OR

- 8 a. Discuss the minimum description length algorithm. (07 Marks)
 b. Apply Naïve Bayes classifier for the below dataset to classify the new instances. (Color = Green, Legs = 2, Height = Tall and Smelly = No)

No.	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

- c. Explain Gibbs algorithm. (07 Marks)
 (06 Marks)

Module-5

- 9 a. Discuss K-Nearest Neighbor learning algorithm. (08 Marks)
 b. Discuss in brief locally weighted linear regression. (06 Marks)
 c. Explain in brief case based reasoning. (06 Marks)

OR

- 10 a. Explain in brief the reinforcement learning technique. (10 Marks)
 b. Discuss the learning tasks and Q learning in the context of reinforcement learning. (10 Marks)

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Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

Big data Analytics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Big data. Explain the classification of Big data. (10 Marks)
- b. Define Scalability and its types along with examples. (10 Marks)

OR

- 2 a. Explain the functions of each layer in Big data architecture design with a diagram. (10 Marks)
- b. Define data preprocessing. Explain in brief the needs of preprocessing. (10 Marks)

Module-2

- 3 a. What is HDFS? Highlight the important design features of the HDFS. (10 Marks)
- b. Bring out the concepts of the HDFS block replication with an example. (10 Marks)

OR

- 4 a. Explain Apache sqoop import and export method with neat diagram. (10 Marks)
- b. Demonstrate any six HBase commands with output. (10 Marks)

Module-3

- 5 a. Explain about NOSQL data store and its characteristics. (10 Marks)
- b. Describe the principle of working of the CAP theorem. (10 Marks)

OR

- 6 a. Demonstrate the working of key-value store with an example. (10 Marks)
- b. Describe the features of MangoDB, and its industrial application. (10 Marks)

Module-4

- 7 a. Describe the Map tasks, Reduce tasks and Map Reduce Execution process. (10 Marks)
- b. Describe the Hive architecture and its characteristics. (10 Marks)

OR

- 8 a. Demonstrate the pig architecture for scripts data flow and processing. (10 Marks)
- b. Differentiate between Pig and Map Reduce, give industrial applications for each. (10 Marks)

Module-5

- 9 a. Explain the simple linear regression analysis. (10 Marks)
- b. Demonstrate frequent item set mining and association rule mining. (10 Marks)

OR

- 10 a. With a neat diagram, write the steps in K-means clustering. (10 Marks)
- b. Explain the purpose of web usage analytics and the significance of web graphs. (10 Marks)

* * * * *

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg, 42+8 = 50, will be treated as malpractice.

CBCS SCHEME

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18CS72

Seventh Semester B.E. Degree Examination, June/July 2023 Big Data and Analytics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define big data, and explain its characteristics. (06 Marks)
- b. List and explain different data sources. (04 Marks)
- c. Explain Big data designing architecture. (10 Marks)

OR

- 2 a. Explain the functions of each of the Big Query layers in big data architecture design using Big Query Cloud Service at Google Cloud Platform. (10 Marks)
- b. (i) Explain big data analytics applications. (05 Marks)
- (ii) Write a short note on data storage and analysis. (05 Marks)

Module-2

- 3 a. With a neat diagram, explain the components of HDFS (Hadoop Distribution File System). (10 Marks)
- b. How does the Hadoop MapReduce data flow work for a word count program? Give an example. (10 Marks)

OR

- 4 a. What is APACHE Flume? Describe the feature components and working of apache flume. (10 Marks)
- b. Explain the features and benefits of apache Hive in hadoop. (10 Marks)

Module-3

- 5 a. Discuss NOSQL data architecture pattern, with an example. (10 Marks)
- b. Explain four different ways for handling big data problems. (10 Marks)

OR

- 6 a. (i) Explain different components of Cassandra. (05 Marks)
- (ii) Explain different data types built into Cassandra. (05 Marks)
- b. (i) Describe different CQL commands and their functionalities. (05 Marks)
- (ii) Write a short note on NOSQL to Manage Big Data. (05 Marks)

Module-4

- 7 a. Describe the significance of apache pig in hadoop. (10 Marks)
- b. With a neat diagram, explain MapReduce Programming model. How does MapReduce enables query processing quickly in Big Data Problems? (10 Marks)

OR

- 8 a. Explain Hive architecture with a neat diagram. (10 Marks)
- b. (i) Differentiate between Pig and MapReduce. (05 Marks)
- (ii) Write a short note on Pig architecture design layers. (05 Marks)

Module-5

- 9 a. What are outliers? Describe the reasons for the presence of outliers in a relationship. (10 Marks)
- b. How can a university student's GPA be predicted from his/her – high school percentage (HSP) of marks? (Assume linear regression) Plot a graph for the same. (10 Marks)

OR

- 10 a. Discuss the different phases of text mining process. (10 Marks)
- b. Write a short note on text mining and web mining. (05 Marks)
- c. Discuss three phases for web usage mining. (05 Marks)

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CBCS SCHEME

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18CS734

Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

User Interface Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define User Interface. Discuss the benefit of good design. (10 Marks)
b. Discuss the characteristics of the Graphical User Interface. (10 Marks)

OR

- 2 a. Compare GUI versus Web Page Design with respect to Devices, Data, Presentation Elements, Navigation, User tasks. (10 Marks)
b. Discuss the following general principles of user interface
i) Aesthetically pleasing ii) Compatibility iii) Consistency
iv) Control v) Simplicity. (10 Marks)

Module-2

- 3 a. List and explain common usability problems in graphical systems. (10 Marks)
b. Discuss any five Human characteristics in interface design. (10 Marks)

OR

- 4 a. Explain Indirect methods of requirement determination in Business Function. (10 Marks)
b. Discuss briefly the guidelines for designing conceptual models. (10 Marks)

Module-3

- 5 a. List and explain different structures of Menus with suitable diagrams. (12 Marks)
b. Discuss Functions and content of Menus. (08 Marks)

OR

- 6 a. Explain the purpose, advantages, disadvantages, guidelines to be followed in designing following menu choices. i) Mark Toggles ii) Toggled Menu Items. (08 Marks)
b. Discuss in detail the following Graphical menus
i) Pull – Down Menu ii) Cascading Menus iii) Popup Menus. (12 Marks)

Module-4

- 7 a. List and discuss in different ways, windows are useful. (12 Marks)
b. Explain filed windows, overlapping windows, and cascading windows their advantages and disadvantages. (08 Marks)

OR

- 8 a. Explain model and modeless and cascading and unfolding windows. (08 Marks)
b. Explain the following with respect to windows
i) frame ii) Window sizing Buttons iii) Scroll bars iv) Split box. (08 Marks)
c. Describe Joystick with advantages and disadvantages. (04 Marks)

Module-5

- 9 a. Explain different command button guide lines. (12 Marks)
b. Describe check boxes, list boxes, palettes with advantages and disadvantages. (08 Marks)

OR

- 10 a. List common presentation controls and discuss any four of them. (08 Marks)
b. Explain the following kinds of tests
i) Cognitive walk through ii) Think-Aloud Evaluations iii) Usability Test. (12 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. 42+8 = 50, will be treated as malpractice.

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18CS734

Seventh Semester B.E. Degree Examination, June/July 2023

User Interface Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define User Interface. Explain its two essential components. (06 Marks)
b. List and explain the characteristics of GUI. (06 Marks)
c. Define object in graphical system. Differentiate between application and data orientation. (08 Marks)

OR

- 2 a. Explain the importance and benefits of good design. (07 Marks)
b. Explain the characteristics of intranet and internet. Also write the differences between them. (08 Marks)
c. Write any five differences between GUI and Web page design. (05 Marks)

Module-2

- 3 a. List and explain the pitfalls in the development path of the design process. (08 Marks)
b. Describe the characteristics of Human Interface Design. (12 Marks)

OR

- 4 a. Explain common usability problem in Web Based Systems. (10 Marks)
b. Explain techniques for determining the user requirement using indirect method. (10 Marks)

Module-3

- 5 a. Explain and illustrate the structure of menus. (10 Marks)
b. Describe the components of web navigation systems with illustration. (10 Marks)

OR

- 6 a. Explain various guidelines to be followed in phrasing of menu. (10 Marks)
b. List all kinds of graphical menus and explain them. (10 Marks)

Module-4

- 7 a. What are the importances of windows? Explain. (12 Marks)
b. Explain any two window management schemes in detail. (08 Marks)

OR

- 8 a. Explain the advantages and disadvantages of frames in web. (10 Marks)
b. Explain the characteristics of the devices touch screen and keyboard. (10 Marks)

Module-5

- 9 a. What are operable controls? Explain the usage of buttons along with their advantages and disadvantages. (10 Marks)
- b. Explain the purpose of prototypes. Discuss any two kinds of prototypes with their importance to the system developers. (10 Marks)

OR

- 10 a. Distinguish between the following :
- (i) Radio buttons and Check boxes.
 - (ii) Tool tips and Ballon tips. (10 Marks)
- b. Explain Cognitive Walkthroughs and Think Aloud evaluation method to perform test in user interface design. (10 Marks)

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Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

Digital Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a neat block diagram, explain the fundamental steps in image processing. (10 Marks)
- b. Explain the concept of sampling and quantization of an image. (05 Marks)
- c. Explain any four fields that use digital image processing. (05 Marks)

OR

- 2 a. Define 4-, 8-, m-adjacency with appropriate examples. (04 Marks)
- b. Explain the various distance measures used with suitable examples. (06 Marks)
- c. Consider the two image subsets S1 and S2 as shown in Fig. Q2 (c) for $V = \{1\}$. Determine whether these two subsets are (i) 4-adjacent (ii) 8-adjacent (iii) M-adjacent

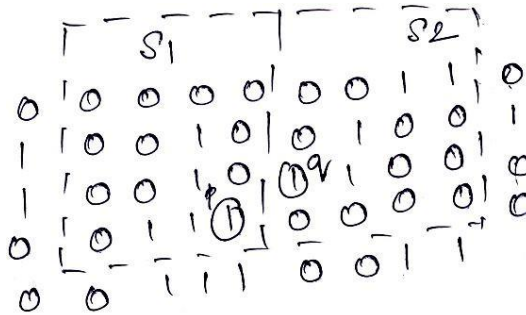


Fig. Q2 (c)

(10 Marks)

Module-2

- 3 a. Explain in detail the basic gray level transformations used in image enhancement. (10 Marks)
- b. Define image enhancement. Explain how arithmetic operators are helpful in image enhancement. (10 Marks)

OR

- 4 a. Define histogram and normalized histogram. Discuss histogram equalization for contrast enhancement. (10 Marks)
- b. Explain image smoothing in spatial domain. (03 Marks)
- c. For a given 4×4 image having gray scales between $[0, 9]$ perform histogram equalization and draw the histogram equalization and draw the histogram of image before and after equalization. 4×4 image is shown in Fig. Q4 (c).

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

Fig. Q4 (c)

(07 Marks)

Module-3

- 5 a. Explain the derivation of DFT from the continuous transform of a sampled function. (10 Marks)
b. Explain homomorphic filter with procedure for applying the same. (10 Marks)

OR

- 6 a. Write the properties of Discrete Fourier transforms. (10 Marks)
b. Discuss Butterworth Low Pass filters and Gaussian Low Pass frequency domain filters. (10 Marks)

Module-4

- 7 a. Explain various Gradient operators used for edge detection. (10 Marks)
b. Explain Global Thresholding using Otsu's method. (10 Marks)

OR

- 8 a. Explain the various approaches used for edge linking. (10 Marks)
b. Explain the basic of intensity thresholding in image segmentation. (10 Marks)

Module-5

- 9 a. Explain in detail the principal types of data redundancies. (10 Marks)
b. Explain Huffman coding technique with an example. (10 Marks)

OR

- 10 a. Explain Lossy and Lossless compression. (10 Marks)
b. Explain Arithmetic coding technique with an example. (10 Marks)

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18CS741

Seventh Semester B.E. Degree Examination, June/July 2023 Digital Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define an image. With block diagram, explain the fundamental steps in digital image processing. (12 Marks)
- b. Explain 4-connectivity, 8-connectivity and m-connectivity with suitable example. (08 Marks)

OR

- 2 a. With necessary diagrams, discuss image digitization process. (10 Marks)
- b. Discuss any four applications of digital image processing. (10 Marks)

Module-2

- 3 a. Define histogram equalization. Develop an algorithm to enhance image quality using this method. (12 Marks)
- b. Discuss any three gray level transformation techniques with suitable graph plots. (08 Marks)

OR

- 4 a. Explain the importance of Sobel and Prewitt operators in image processing. (08 Marks)
- b. Discuss Laplacian and Log Edge detectors with suitable mathematical model. (12 Marks)

Module-3

- 5 a. List the properties of DFT. Explain Discrete Fourier Transform Process along with its computational complexity. (12 Marks)
- b. Discuss steps in frequency domain based filtering techniques. (08 Marks)

OR

- 6 a. Discuss the filters used to smooth an image in frequency domain. (10 Marks)
- b. Explain the significance of homomorphic filters with a block diagram. (10 Marks)

Module-4

- 7 a. Define image segmentation. What are the properties of segmentation? (10 Marks)
- b. Explain 'Region Growing' and 'Split and Merge' approach used in image segmentation. (10 Marks)

OR

- 8 a. Explain how Hough Transform helps in extracting line segments from an image. (10 Marks)
- b. Define thresholding. Explain Global thresholding technique with an example. (10 Marks)

Module-5

- 9 a. Define image compression. Explain general image compression model with a block diagram. (10 Marks)
- b. Develop an algorithm that encodes data using Huffman coding technique. Illustrate Huffman coding for the following data:

Data	S ₁	S ₂	S ₃	S ₄	S ₅	S ₆
Probability	0.1	0.4	0.06	0.1	0.04	0.3

(10 Marks)

OR

- 10 a. Discuss LZW encoding and decoding technique with an example. (12 Marks)
- b. With a block, discuss transform coding technique. (08 Marks)

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Seventh Semester B.E. Degree Examination, June/July 2023

Python Application Programming

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1
 - a. List the rules to declare variable in python and salient features of python. (06 Marks)
 - b. Write a python program to find biggest of three numbers accepted from the user. (06 Marks)
 - c. Explain concept of type conversion function and math functions in python with examples. (08 Marks)

OR

- 2
 - a. Explain break and continue statement with example in python. (06 Marks)
 - b. How python handles the exceptions? Explain with an example program. (06 Marks)
 - c. What are user defined functions? Write a function compute grade that takes a score as its parameter and returns a grade as string.

Score	Grade
> = 0.9	A
> = 0.8	B
> = 0.7	C
> = 0.6	D
> = 0.6	F

(08 Marks)

Module-2

- 3
 - a. Discuss string handling methods in python with examples. (08 Marks)
 - b. Write a python program to demonstrate counting, summing and average of elements using loops. (06 Marks)
 - c. Write a python program to check whether given string is palindrome or not. (06 Marks)

OR

- 4
 - a. Explain for loop and while loop and write a program to generate Fibonacci series up to the given limit. (08 Marks)
 - b. Write a python program to count the number of occurrences of a word in a given file. (06 Marks)
 - c. Write a program to read through a file and look for lines of the form.
X-DSPAM – confidence : 0.8475
When you encounter line that starts with “X – DSPAM confidence” pull apart the line to extract the floating points numbers on the line, count there line and then compute the total of the spam confidence. (06 Marks)

Module-3

- 5
 - a. What are lists? Describe list methods in python. Write a program to find biggest element in the list. (08 Marks)
 - b. How tuples are created in python? Explain different ways of accruing and creating them. (06 Marks)
 - c. Write python program to search for lines that start with ‘F’ followed by 2 characters, followed by ‘m’ and print there lines. (06 Marks)

OR

- 6 a. What is dictionary? Write a python program to count occurrence of words in a file and print the count. (08 Marks)
- b. Explain extracting data using regular expressions implement python program to find for lines having '@' symbol between non blank characters in a file. (08 Marks)
- c. Compare and contrast lists and tuples. (04 Marks)

Module-4

- 7 a. Explain init and str method with an example python program. (10 Marks)
- b. What is Operator overloading? Write python code to overload "+", "-", "*" and "/" operators by providing the methods __add__, __sub__, __mul__ and __div__. (06 Marks)
- c. Define class and object. Explain with example. (04 Marks)

OR

- 8 a. What is a pure function? Write a python program to find duration of event if start and end time is given by defining class TIME. (10 Marks)
- b. Write a python program that uses date time module within a class takes birthday as input and points the age and the no of days, hours, minutes and seconds until next birthday. (10 Marks)

Module-5

- 9 a. Demonstrate with the help of python construct
- How to retrieve an image over HTTP
 - How to retrieve web pages with Urllib. (10 Marks)
- b. Write a note on Google Geocoding Web service. Using python supported libraries demonstrate with snippet code. (10 Marks)

OR

- 10 a. Write a python program for creating employee database, inserting records and selecting employers working in the company. (10 Marks)
- b. Describe creation of database table using database cursor architecture. (10 Marks)

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Third Semester B.E. Degree Examination, Jan./Feb. 2023
Transform Calculus, Fourier Series and Numerical
Techniques

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Find the Laplace transform of $te^{2t} \frac{2 \sin 3t}{t}$. (06 Marks)

- b. Given that $f(t) = \begin{cases} B, & 0 < t < \frac{\gamma}{2} \\ E, & \frac{\gamma}{2} < t < a \end{cases}$

where $f(t+a) \equiv f(t)$ show that $L\{f(t)\} = \frac{E}{s} \tan h\left(\frac{\pi s}{4}\right)$. (07 Marks)

- c. Using convolution theorem obtain the inverse Laplace transform of the following function:

$$\frac{(s-1)(s^2+1)}{(s-1)(s^2+1)}$$

(07 Marks)

OR

- 2 a. Find the inverse Laplace transform of:

$$\frac{s+5}{s^2-6s+13}$$

(06 Marks)

- b. Express the following function in terms of unit step function and hence find their Laplace transform.

$$f(t) = \begin{cases} 1, & 0 < t < 1 \\ t^2, & 1 < t \leq 2 \\ t^2, & t > 2. \end{cases}$$

(07 Marks)

- c. Solve the following initial value problem by using Laplace transform:

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = e^{-t}, \quad y(0) = 0, \quad y'(0) = 0$$

(07 Marks)

Module-2

- 3 a. Obtain Fourier series of $f(x) = \frac{\pi-x}{2}$ in $0 < x < 2\pi$. Hence deduce that

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$$

(06 Marks)

- b. Find a cosine Fourier series for $f(x) = (x-1)^2$, $0 \leq x \leq 1$.

(07 Marks)

- c. Obtain the Fourier series of y upto the first harmonic for the following values.

x°	45	90	135	180	225	270	315	360
y	4.0	3.8	2.4	2.0	-1.5	0	2.8	3.4

(07 Marks)

1 of 3

OR

- 4 a. Obtain Fourier series for

$$f(x) = \begin{cases} \pi x & \text{in } 0 \leq x \leq 1 \\ \pi(2-x) & \text{in } 1 \leq x \leq 2 \end{cases}$$

(06 Marks)

- b. Obtain the sine half range series for the function:

$$f(x) = 1 - \left(\frac{x}{\pi}\right) \ln \quad 0 \leq x \leq \pi.$$

(07 Marks)

- c. The following values of y and x are given. Find Fourier series of upto first harmonics.

x	0	2	4	6	8	10	12
y	9.0	18.2	24.4	27.8	27.5	22.0	9.0

(07 Marks)

Module-3

- 5 a. If $f(x) = \begin{cases} 1-x^2, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}$. Find Fourier transform of $f(x)$ and hence find the value of

$$\int_0^\infty x \cos x \cdot e^{-\sin x} dx.$$

(06 Marks)

- b. Find the Fourier sine transform of $f(x) = e^{-|x|}$ and hence evaluate

$$\int_0^\infty \frac{x \sin mx}{1+x^2} dx, \quad m > 0.$$

(07 Marks)

- c. Solve by using Z-Transforms $U_{n+2} + 2U_{n+1} + U_n = n$ with $U_0 = 0 = U_1$.

(07 Marks)

OR

- 6 a. Obtain the Fourier cosine transform of the function:

$$f(x) = \begin{cases} 4x, & 0 < x < 1 \\ 4-x, & 1 < x \leq 4 \\ 0, & x > 4. \end{cases}$$

(06 Marks)

- b. Obtain the Z-transform of $\cos n\theta$ and $\sin n\theta$.

(07 Marks)

- c. Compute the inverse Z-transform of $\frac{3z^2-2z}{(5z-1)(5z+2)}$.

(07 Marks)

Module-4

- 7 a. Classify the following partial differential equations:

i) $x^2 u_{xx} + (1-y^2) u_{yy} = 0$, $-\infty < x < \infty$, $-1 < y < 1$

ii) $(1+x^2) u_{xx} + (5+2x^2) u_x + (4+x^2) u_{yy} = 0$

iii) $(x+1) u_{xx} - 2(x+2) u_{xy} + (x+3) u_{yy} = 0$.

(10 Marks)

- b. Solve $u_t = u_{xx}$ subject to the conditions $u(0, t) = 0 = u(1, t)$ and $u(x, 0) = \sin \pi x$ by taking $h = 0.2$ for 5 levels. Further write down the following values from the table

i) $u(0.2, 0.04)$
ii) $u(0.4, 0.08)$
iii) $u(0.6m, 0.06)$.

(10 Marks)

2 of 3

OR

8. a. Solve the elliptic equation $u_{xx} + u_{yy} = 0$ for the following square Mesh with boundary values as shown. Find the iterative values of u_4 (to 9) to the nearest integer.



Fig.Q8(a)

- b. Solve $25u_x = u_t$ at the pivotal points given $u(0, t) = 0 = u(5, t)$, $u(x, 0) = 0$ and $u(x, 0) = \begin{cases} 20x, & 0 \leq x \leq 1 \\ 5(5-x), & 1 \leq x \leq 5 \end{cases}$ by taking $h = 1$ compute $u(x, t)$ for $0 \leq t \leq 1$. (10 Marks)

Module-5

9. a. Given $y'' - xy' - y = 0$ with the initial conditions $y(0) = 1, y'(0) = 0$ compute $y(0.2)$ using fourth order Runge-Kutta method. (06 Marks)
 b. Derive the Euler's equation. (07 Marks)
 c. Find the extremal of the functional $\int_{x_1}^{x_2} (y'^2 + y^2 + 2ye^x) dx$. (07 Marks)

OR

10. a. Obtain the solution of the equation $2 \frac{d^2 y}{dx^2} = 4x + \frac{dy}{dx}$ by computing the value of $y(1.4)$ by applying Milne's method using following data :

x	1	1.1	1.2	1.3
y	2	2.2156	2.4649	2.7514
y'	2	2.3178	2.6725	3.0657

(06 Marks)

- b. Find the curve for which the functional $\int_0^1 [y'^2 + 12xy] dx$ with $y(0) = 0$ and $y(1) = 1$ can be determined. (07 Marks)
 c. Prove that the shortest distance between two points in a plane is straight line. (07 Marks)

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21CS32

Third Semester B.E. Degree Examination, Jan./Feb. 2023 Data Structures and Applications

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is linear array? Discuss the representation of linear array in memory. (06 Marks)
- b. Differentiate between static and dynamic memory allocations. Discuss four dynamic memory allocation functions. (06 Marks)
- c. Write a menu driven program in C for the following array operations:
- (i) Inserting an element (ELEM) at a given valid position.
 - (ii) Deleting an element at a given valid position.
 - (iii) Display of array elements.
 - (iv) Exit
- Support the program with functions for each of the above operations. (08 Marks)

OR

- 2 a. Give Abstract Data Type (ADT) for arrays. How array can be declared and initialized? (06 Marks)
- b. With suitable example, discuss self-referential structures. (06 Marks)
- c. Define Sparse matrix. How to represent a Sparse matrix? Write an algorithm/function to transpose a given Sparse matrix. (08 Marks)

Module-2

- 3 a. Define Stack. Discuss how to represent stack using dynamic arrays. (06 Marks)
- b. Write a menu driven C program for the following operations on STACK of integers:
- (i) Push an element on to stack
 - (ii) Pop an element from the stack
 - (iii) Display the content of stack
 - (iv) Exit
- Show the overflow and underflow conditions. (06 Marks)
- c. What are the disadvantages of ordinary queue? Discuss the implementation of circular queue using arrays. (08 Marks)

OR

- 4 a. What is Recursion? Write recursive function to solve Towers of Hanoi problem. (06 Marks)
- b. Discuss the following:
- (i) Double Ended Queue
 - (ii) Priority Queue
- (06 Marks)
- c. Write an algorithm to convert infix expression to postfix expression. Show the content of stack to convert the following infix expression:
- $A * (B + D) / E - F * (G + H / K)$ (08 Marks)

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. $42+8=50$, will be treated as malpractice.

Module-3

- 5 a. Write a C function to concatenate two singly linked list. (06 Marks)
 b. Give the structure definition for singly linked list. Write a C function to:
 (i) Insert an element at the end
 (ii) Delete a node at the beginning (08 Marks)
 c. Discuss how to read a polynomial consisting of 'n' terms implemented using linked list. (06 Marks)

OR

- 6 a. Write a function to delete a node whose information field is specified in singly linked list. (06 Marks)
 b. What is circular doubly linked list? Write a C function to perform the following operations on circular doubly linked list:
 (i) Insert a node at the beginning
 (ii) Delete a node from the list (08 Marks)
 c. Discuss how to implement stacks and queues using linked list. (06 Marks)

Module-4

- 7 a. Define binary tree. List and discuss any two properties of binary tree. (06 Marks)
 b. Write a function to perform the following operations on Binary Search Tree (BST):
 (i) Deletion from a BST
 (ii) Inserting an element into a BST (08 Marks)
 c. Define Threaded Binary Tree. Discuss In-threaded binary tree. (06 Marks)

OR

- 8 a. Discuss how binary tree are represented using (i) Array (ii) Linked list (06 Marks)
 b. Discuss inorder, preorder, postorder and level order traversal with suitable recursive function for each. (08 Marks)
 c. Write a C function to evaluate an expression using expression tree. (06 Marks)

Module-5

- 9 a. Design a C program for the following operation on Graph (G) of cities:
 (i) Create a graph of N cities using adjacency matrix
 (ii) Print all the nodes reachable from a given starting node in a digraph using BFS/DFS method (10 Marks)
 b. Discuss AVL tree with an example. Write a function for insertion into an AVL tree. (10 Marks)

OR

- 10 a. Define hashing. What are the two criteria, a good hash function should satisfy? Discuss open addressing and chaining method with an example. (10 Marks)
 b. Define Red-Black tree, Splay tree and B tree. Discuss the method to insert an element into Red-Black tree. (10 Marks)

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Third Semester B.E. Degree Examination, Jan./Feb. 2023

Analog and Digital Electronics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.**Module-1**

- 1 a. Derive an expression for collector current and collector emitter voltage of voltage divider bias circuit (accurate analysis). (08 Marks)
- b. Explain relaxation oscillator. (06 Marks)
- c. Sketch and explain the working of Peak detector. (06 Marks)

OR

- 2 a. Explain R-2R ladder type DAC with a neat diagram. (06 Marks)
- b. List the advantages of active filters over passive filters. (06 Marks)
- c. For the circuit shown in Fig. Q2 (c) below find the value of R_1 and R_2 if supply voltages are +12 and -12 V. Assume hysteresis with -6 V.

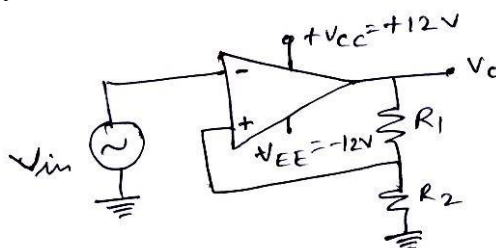


Fig. Q2 (c)

(08 Marks)

Module-2

- 3 a. Find all the prime implicants of the function,
 $f(a, b, c, d) = \Pi(0, 2, 3, 4, 5, 12, 13) + \Pi d(8, 10)$
 using the Quine-McCluskey method. (10 Marks)
- b. Plot the Karnaugh maps and find all the minimal sums and minimal products of the following Boolean functions.
 (i) $f(a, b, c) = \sum(2, 4, 5, 6, 7)$
 (ii) $f(a, b, c) = \Pi(1, 4, 5, 6)$ (10 Marks)

OR

- 4 a. With an example, explain Petrik's method. (06 Marks)
- b. For the given Boolean function, determine a minimal sum and a minimal product using MEV techniques using a, b and c as the map variables.
 $f = \sum(3, 4, 5, 7, 8, 11, 12, 13, 15)$ (08 Marks)
- c. Explain Entered variable map method. (06 Marks)

Module-3

- 5 a. Explain the importance of three-state buffer. (06 Marks)
 b. With a neat diagram, explain 3 to 8 line decoder. (06 Marks)
 c. What is a multiplexer? Write the logic diagram for 8 : 1 multiplexer using 4 input AND and OR gates. (08 Marks)

OR

- 6 a. Discuss different types of hazards in combinational circuits. (08 Marks)
 b. Distinguish between combinational and sequential circuit. (06 Marks)
 c. Write a note on PLA and PAL. (06 Marks)

Module-4

- 7 a. Explain the working of JK master slave flip-flop with a sketch, truth table and symbol. (06 Marks)
 b. What is D flip flop? Illustrate the operation of the clear and preset inputs in D-flip-flop with timing diagram. (08 Marks)
 c. What is VHDL? Show how to model the 4 to 1 multiplexer using a VHDL conditional assignment statement. (06 Marks)

OR

- 8 a. What is T-flip-flop? Show how to convert D-flip flop into T-flip-flop. (08 Marks)
 b. What are the three different models for writing a module body in VHDL? Give example for any one model. (06 Marks)
 c. Explain with a neat diagram, VHDL program structure. (06 Marks)

Module-5

- 9 a. With a neat diagram, explain 4-bit parallel adder with accumulator. (10 Marks)
 b. Define counter. Design mod-5 counter using J-K flip flop. (10 Marks)

OR

- 10 a. With neat diagram, explain 4 bit SISO register. (08 Marks)
 b. Mention the Application of shift registers. (05 Marks)
 c. Explain the working of a 3 bit shift register. (07 Marks)

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CBCS SCHEME

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Third Semester B.E. Degree Examination, Jan./Feb. 2023 Computer Organization and Architecture

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With the help of a neat block diagram discuss the basic operational concept of a computer. (08 Marks)
- b. Write a program to evaluate the arithmetic statement $Y = (A + B) * (C + D)$ using three address, two address, one address and zero address instruction. (08 Marks)
- c. Write the basic performance equation indicate the role of each parameter in the equation. (04 Marks)

OR

- 2 a. Define Addressing Mode. Explain the various addressing mode. (10 Marks)
- b. With proper example explain Big - Endian and Little - Endian of byte addressing. (06 Marks)
- c. What is performance measurement? Explain the overall SPEC rating of a computer. (04 Marks)

Module-2

- 3 a. With respect to handling interrupts from multiple devices explain:
(i) Interrupt nesting (ii) Dairy chain method. (10 Marks)
- b. What is Bus arbitration? Explain centralized and distributed arbitration method with neat diagrams. (10 Marks)

OR

- 4 a. Illustrate a program that reads one line from keyboard, stored it in memory buffer and echoes it back to display in I/O interfaces. (10 Marks)
- b. Discuss with a neat circuit diagram, the general 8 bit parallel interface circuit. (10 Marks)

Module-3

- 5 a. Explain the internal organization of 16-megabit DRAM chip configured as $2M \times 8$. (08 Marks)
- b. With a neat figure illustrate the structure of synchronous DRAM (SDRAM). (08 Marks)
- c. Discuss about any two types of Read Only Memory (ROM). (04 Marks)

OR

- 6 a. State the importance of cache memory and describe the different types of cache mapping techniques with diagram. (12 Marks)
- b. With relevant figure explain organization of $(1k \times 1)$ memory chip. (08 Marks)

Module-4

- 7 a. With the help of logic diagram explain 4-bit carry look adder and its operation. (10 Marks)
- b. Illustrate the hardware arrangement for sequential multiplication with an example. (10 Marks)

OR

- 8 a. Draw the single bus architecture and explain the control sequence for execution of instruction ADD (R3), R1. (10 Marks)
b. With neat sketches, explain the detailed organization of hardwired control unit. (10 Marks)

Module-5

- 9 a. With a suitable example explain the concept of pipeline processing. (10 Marks)
b. Draw and explain pipeline for floating point addition and subtraction. (10 Marks)

OR

- 10 a. With the help of flowchart and timing diagram explain four segment instruction pipeline. (10 Marks)
b. Explain the organization of SIMD array processor with an appropriate diagram. (10 Marks)

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Question Paper Version : A

Third Semester B.E./B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Fill in the blank by translating the given English word to Kannada :

1. You : _____
a) neenu b) Naanu c) yaaru d) Ivaru
2. His : _____
a) NeeVu b) avaru c) aVaLu d) avana
3. What : _____
a) idu b) adu c) yenu d) yelli
4. Friend : _____
a) Gumpu b) Maava c) Sneheta d) GurugaLu
5. Drama : _____
a) naataka b) Kate c) haaDu d) Yaavadu alla
6. Choose the Mathematical + Symbol in Kannada
a) Kale b) Koodu c) Jodisu d) Bhagisu
7. Country in Kannada will be called as
a) D'weshu b) Vesha c) Desha d) Vishesha

Note : Substitute the word from the following each sentence in appropriate places :

8. idu : idakke then adu _____
a) adakke b) idakke c) Horakke d) Horakkee
9. Uuru : Urige then avaru _____
a) ivarige b) avarige c) Nimage d) Nawage

10. Mane : maneyalli then Pustaka
a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli
11. Raja : Rajanaddu, then Raani _____
a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu
- Note : Write the English word for the given Kannada word :**
12. HaNNu : _____
a) seeds b) flower c) Tree d) Fruit
13. pramaaNu : _____
a) promise b) very good c) purpse d) Thankyou
14. tangi : _____
a) Aunti b) Uncle c) Sister d) Mother
15. gaNita : _____
a) Biology b) Physics c) English d) Mathematics
16. biLi : _____
a) red b) Black c) White d) Green
17. doDDa : _____
a) Big b) Small c) fat d) thin
18. Aarogya : _____
a) Tomarrow b) Health c) wealth d) cold
19. gurugaLu
a) Former b) Teacher c) Father d) Uncle
20. uuTa : _____
a) Meals b) Drinks c) Get up d) breakfast
21. KoDabeDa : _____
a) Give me b) Give us c) give d) Don't give
22. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair
23. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
24. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
25. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep
26. huTTu : _____
a) Good b) die c) Birth d) bed

27. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
28. daKShiNa : _____
a) West b) East c) South d) North
29. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
30. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
31. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
32. granthalaya : _____
a) House b) office c) college d) Library
33. baNNa : _____
a) White b) Banana c) Colour d) Red

Note : Translate the English question to Kannada :

34. Where is he?
a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
35. Where shall we meet?
a) NaVu yaaru? b) Sanje barattiya?
c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
36. When is the physics class?
a) physics class yaavagaa? b) Sanje physicedeyan?
c) physic Lab elli ede? d) yaavadu alla
37. When are you going to College?
a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
c) neenu yavagu hoguttiya? d) enu MaDuttiddiya?
38. Had your breakfast?
a) YaaVagu Tindi? b) TinDi ayitaa?
c) oota ayta? d) TinDi yestu beku?

Note : Translate the Question Kannada to English :

39. eshTu ganTe Saar?
a) Can I go Sir? b) will I come Sir?
c) 8 Clock Sir d) What is the time Sir?
40. Ee pustaKa yaaraduu?
a) Is that book yours b) whose book is this?
c) This book belongs to me. d) none

41. ethaKKe hogidde?
 a) Why did you go? b) Where did you go?
 c) When you are going? d) None of these
42. nimma Kalejinaa hesarenu?
 a) What is your college name? b) Where is your college
 c) How old is your college? d) Which college are you studying?
43. naanu yenu MaDabhahudu?
 a) What did you do? b) What should we do?
 c) What can I do? d) None
44. eegale baa
 a) come immediatly b) Go immediatly c) come Tomorrow d) None
45. dayamaDi niiVu oLagaDe banni
 a) When you come b) Come inside
 c) Don't come hear d) you please come inside
46. Kallegige raje KoTTi ddare
 a) Holiday is given to college b) Don't come to college
 c) Today college is there d) None
47. Raamanu nanaginta doDDavanu
 a) Rama is older than me b) Rama is elder than me
 c) Rama is my friend d) None
48. VandneguLu Saar
 a) Very good b) Thank you sir
 c) May I come Sir d) Good morning Sir
49. baalehaNNu
 a) Mango b) Chicku c) Banana d) Orange
50. ToTa _____
 a) Room b) Bangle c) House d) Garden

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Question Paper Version : B

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note : Translate the English question to Kannada :

1. Where is he?
a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
2. Where shall we meet?
a) NaVu yaaru? b) Sanje barattiya?
c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
3. When is the physics class?
a) physics class yaavagaa? b) Sanje physicedeyan?
c) physic Lab elli ede? d) yaavadu alla
4. When are you going to College?
a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
c) neenu yavagu hogutitya? d) enu MaDuttiddiya?
5. Had your breakfast?
a) YaaVagu Tindi? b) TinDi ayitaa?
c) oota ayta? d) TinDi yestu beku?

Note : Write the English word for the given Kannada word :

6. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
7. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
8. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep

9. huTTu : _____
a) Good b) die c) Birth d) bed
10. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
11. daKShiNa : _____
a) West b) East c) South d) North
12. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
13. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
14. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
15. granthalaya : _____
a) House b) office c) college d) Library
16. baNNa : _____
a) White b) Banana c) Colour d) Red
17. HaNNu : _____
a) seeds b) flower c) Tree d) Fruit
18. pramaaNa : _____
a) promise b) very good c) purpse d) Thankyou
19. tangi : _____
a) Aunti b) Uncle c) Sister d) Mother
20. gaNita : _____
a) Biology b) Physics c) English d) Mathematics
21. biLi : _____
a) red b) Black c) White d) Green
22. doDDa : _____
a) Big b) Small c) fat d) thin
23. Aarogya : _____
a) Tomarrow b) Health c) wealth d) cold
24. gurugaLu
a) Former b) Teacher c) Father d) Uncle
25. uuTa : _____
a) Meals b) Drinks c) Get up d) breakfast

26. KoDabeDa : _____
a) Give me b) Give us c) give d) Don't give

27. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair

Note : Substitute the word from the following each sentence in appropriate places :

28. idu : idakke then adu _____
a) adakke b) idakke c) Horakke d) Horakkee

29. UUru : Urige then avaru _____
a) ivarige b) avarige c) Nimage d) Nawage

30. Mane : maneyalli then Pustaka
a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli

31. Raja : Rajanaddu, then Raani _____
a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

Note : Translate the Question Kannada to English :

32. eshTu ganTe Saar?
a) Can I go Sir? b) will I come Sir?
c) 8 Clock Sir d) What is the time Sir?

33. Ee pustaKa yaaraduu?
a) Is that book yours b) whose book is this?
c) This book belongs to me. d) none

34. ethaKKe hogidde?
a) Why did you go? b) Where did you go?
c) When you are going? d) None of these

35. nimma Kalejinaa hesarenu?
a) What is your college name? b) Where is your college
c) How old is your college? d) Which college are you studying?

36. naanu yenu MaDabhahudu?
a) What did you do? b) What should we do?
c) What can I do? d) None

37. eegale baa
a) come immediatly b) Go immediatly c) come Tomorrow d) None

38. dayamaDi niiVu oLagaDe banni
a) When you come b) Come inside
c) Don't come hear d) you please come inside

39. Kallegige raje KoTTi ddare
a) Holiday is given to college b) Don't come to college
c) Today college is there d) None

40. Raamanu nanaginta doDDavanu
 a) Rama is older than me
 b) Rama is elder than me
 c) Rama is my friend
 d) None
41. VandneguLu Saar
 a) Very good
 b) Thank you sir
 c) May I come Sir
 d) Good morning Sir
42. baalehaNNu
 a) Mango
 b) Chicku
 c) Banana
 d) Orange
43. ToTa _____
 a) Room
 b) Bangle
 c) House
 d) Garden

Fill in the blank by translating the given English word to Kannada :

44. You : _____
 a) neenu
 b) Naanu
 c) yaaru
 d) Ivaru
45. His : _____
 a) NeeVu
 b) avaru
 c) aVaLu
 d) avana
46. What : _____
 a) idu
 b) adu
 c) yenu
 d) yelli
47. Friend : _____
 a) Gumpu
 b) Maava
 c) Sneheta
 d) GurugaLu
48. Drama : _____
 a) naataka
 b) Kate
 c) haaDu
 d) Yaavadu alla
49. Choose the Mathematical + Symbol in Kannada
 a) Kale
 b) Koodu
 c) Jodisu
 d) Bhagisu
50. Country in Kannada will be called as
 a) D'weshu
 b) Vesha
 c) Desha
 d) Vishesha

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Question Paper Version : C

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
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3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note : Write the English word for the given Kannada word :

1. HaNNu : _____
a) seeds b) flower c) Tree d) Fruit
2. pramaaNa : _____
a) promise b) very good c) purpse d) Thankyou
3. tangi : _____
a) Aunti b) Uncle c) Sister d) Mother
4. gaNita : _____
a) Biology b) Physics c) English d) Mathematics
5. biLi : _____
a) red b) Black c) White d) Green
6. doDDa : _____
a) Big b) Small c) fat d) thin
7. Aarogya : _____
a) Tomarrow b) Health c) wealth d) cold
8. gurugaLu
a) Former b) Teacher c) Father d) Uncle
9. uuTa : _____
a) Meals b) Drinks c) Get up d) breakfast

10. KoDabeDa : _____
a) Give me b) Give us c) give d) Don't give
11. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair
12. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
13. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
14. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep
15. huTTu : _____
a) Good b) die c) Birth d) bed
16. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
17. daKShiNa : _____
a) West b) East c) South d) North
18. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
19. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
20. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
21. granthalaya : _____
a) House b) office c) college d) Library
22. baNNa : _____
a) White b) Banana c) Colour d) Red

Note : Translate the Question Kannada to English :

23. eshTu ganTe Saar?
a) Can I go Sir? b) will I come Sir?
c) 8 Clock Sir d) What is the time Sir?
24. Ee pustaKa yaaraduu?
a) Is that book yours b) whose book is this?
c) This book belongs to me. d) none
25. ethaKKe hogidde?
a) Why did you go? b) Where did you go?
c) When you are going? d) None of these

26. nimma Kalejinaa hesarenu?
 a) What is your college name? b) Where is your college
 c) How old is your college? d) Which college are you studying?
27. naanu yenu MaDabhahudu?
 a) What did you do? b) What should we do?
 c) What can I do? d) None
28. eegale baa
 a) come immediatly b) Go immediatly c) come Tomorrow d) None
29. dayamaDi niiVu oLagaDe banni
 a) When you come b) Come inside
 c) Don't come hear d) you please come inside
30. Kallegige raje KoTTi ddare
 a) Holiday is given to college b) Don't come to college
 c) Today college is there d) None
31. Raamanu nanaginta doDDavanu
 a) Rama is older than me b) Rama is elder than me
 c) Rama is my friend d) None
32. VandneguLu Saar
 a) Very good b) Thank you sir
 c) May I come Sir d) Good morning Sir
33. baalehaNNu
 a) Mango b) Chicku c) Banana d) Orange
34. ToTa _____
 a) Room b) Bangle c) House d) Garden
- Fill in the blank by translating the given English word to Kannada :**
35. You : _____
 a) neenu b) Naanu c) yaaru d) Ivaru
36. His : _____
 a) NeeVu b) avaru c) aVaLu d) avana
37. What : _____
 a) idu b) adu c) yenu d) yelli
38. Friend : _____
 a) Gumpu b) Maava c) Sneheta d) GurugaLu
39. Drama : _____
 a) naataka b) Kate c) haaDu d) Yaavadu alla

40. Choose the Mathematical + Symbol in Kannada
 a) Kale b) Koodu c) Jodisu d) Bhagisu
41. Country in Kannada will be called as
 a) D'weshu b) Vesha c) Desha d) Vishesha

Note : Translate the English question to Kannada :

42. Where is he?
 a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
43. Where shall we meet?
 a) NaVu yaaru? b) Sanje barattiya?
 c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
44. When is the physics class?
 a) physics class yaavagaa? b) Sanje physicedeyan?
 c) physic Lab elli ede? d) yaavadu alla
45. When are you going to College?
 a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
 c) neenu yavagu hogutitya? d) enu MaDuttiddiya?
46. Had your breakfast?
 a) YaaVagu Tindi? b) TinDi ayitaa?
 c) oota ayta? d) TinDi yestu beku?

Note : Substitute the word from the following each sentence in appropriate places :

47. idu : idakke then adu _____
 a) adakke b) idakke c) Horakke d) Horakkee
48. UUru : Urige then avaru _____
 a) ivarige b) avarige c) Nimage d) Nawage
49. Mane : maneyalli then Pustaka _____
 a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli
50. Raja : Rajanaddu, then Raani _____
 a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note : Translate the Question Kannada to English :

1. eshTu ganTe Saar?
a) Can I go Sir?
b) will I come Sir?
c) 8 Clock Sir
d) What is the time Sir?
2. Ee pustaKa yaaraduu?
a) Is that book yours
b) whose book is this?
c) This book belongs to me.
d) none
3. ethaKKe hogidde?
a) Why did you go?
b) Where did you go?
c) When you are going?
d) None of these
4. nimma Kalejinaa hesarenu?
a) What is your college name?
b) Where is your college
c) How old is your college?
d) Which college are you studying?
5. naanu yenu MaDabhahudu?
a) What did you do?
b) What should we do?
c) What can I do?
d) None
6. eegale baa
a) come immediatly
b) Go immediatly
c) come Tomorrow
d) None
7. dayamaDi niiVu oLagaDe banni
a) When you come
b) Come inside
c) Don't come hear
d) you please come inside

8. Kallegige raje KoTTi ddare
a) Holiday is given to college b) Don't come to college
c) Today college is there d) None
9. Raamanu nanaginta doDDavanu
a) Rama is older than me b) Rama is elder than me
c) Rama is my friend d) None
10. VandneguLu Saar
a) Very good b) Thank you sir
c) May I come Sir d) Good morning Sir
11. baalehaNNu
a) Mango b) Chicku c) Banana d) Orange
12. ToTa _____
a) Room b) Bangle c) House d) Garden

Fill in the blank by translating the given English word to Kannada :

13. You : _____
a) neenu b) Naanu c) yaaru d) Ivaru
14. His : _____
a) NeeVu b) avaru c) aVaLu d) avana
15. What : _____
a) idu b) adu c) yenu d) yelli
16. Friend : _____
a) Gumpu b) Maava c) Sneheta d) GurugaLu
17. Drama : _____
a) naataka b) Kate c) haaDu d) Yaavadu alla
18. Choose the Mathematical + Symbol in Kannada
a) Kale b) Koodu c) Jodisu d) Bhagisu
19. Country in Kannada will be called as
a) D'wsha b) Vesha c) Desha d) Vishsha

Note : Translate the English question to Kannada :

20. Where is he?
a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
21. Where shall we meet?
a) NaVu yaaru? b) Sanje barattiya?
c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
22. When is the physics class?
a) physics class yaavagaa? b) Sanje physicedeyan?
c) physic Lab elli ede? d) yaavadu alla

23. When are you going to College?
 a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
 c) neenu yavagu hogutitya? d) enu MaDuttiddiya?
24. Had your breakfast?
 a) YaaVagu Tindi? b) TinDi ayitaa?
 c) oota ayta? d) TinDi yestu beku?

Note : Substitute the word from the following each sentence in appropriate places :

25. idu : idakke then adu _____
 a) adakke b) idakke c) Horakke d) Horakkee
26. UURu : Urige then avaru _____
 a) ivarige b) avarige c) Nimage d) Nawage
27. Mane : maneyalli then Pustaka
 a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli
28. Raja : Rajanaddu, then Raani _____
 a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

Note : Write the English word for the given Kannada word :

29. HaNNU : _____
 a) seeds b) flower c) Tree d) Fruit
30. pramaaNu : _____
 a) promise b) very good c) purpse d) Thankyou
31. tangi : _____
 a) Aunti b) Uncle c) Sister d) Mother
32. gaNita : _____
 a) Biology b) Physics c) English d) Mathematics
33. biLi : _____
 a) red b) Black c) White d) Green
34. doDDa : _____
 a) Big b) Small c) fat d) thin
35. Aarogya : _____
 a) Tomarrow b) Health c) wealth d) cold
36. gurugaLu
 a) Former b) Teacher c) Father d) Uncle
37. uuTa : _____
 a) Meals b) Drinks c) Get up d) breakfast
38. KoDabeDa : _____
 a) Give me b) Give us c) give d) Don't give

39. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair
40. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
41. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
42. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep
43. huTTu : _____
a) Good b) die c) Birth d) bed
44. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
45. daKShiNa : _____
a) West b) East c) South d) North
46. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
47. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
48. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
49. granthalaya : _____
a) House b) office c) college d) Library
50. baNNa : _____
a) White b) Banana c) Colour d) Red

USN

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Question Paper Version : A

Third Semester B.E./B.Tech Degree Examination, Jan./Feb. 2023
Samskruthika Kannada

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners on the OMR sheets are strictly prohibited.**

ಸಂಕೇತಿಸಿದ ಉತ್ತರಗಳನ್ನು ಹೀಗೆ ಉತ್ತರಿಸಿ :

1. 9983fi6 ಲಿಪಿ ಲಿಪಿ ಸಂಖ್ಯೆ 23. 83ಯ ಲಿಪಿ ಲಿಪಿ ಸಂಖ್ಯೆ?
 - a) ಕೃಷ್ಣಾಕ್ಷರಿಸಂಖ್ಯೆ
 - b) ಬುಕ್ಕಾಕ್ಷರಿಸಂಖ್ಯೆ
 - c) ಹಕ್ಕಬುಕ್ಕ
 - d) ಜಿ'ರಿಸಂಖ್ಯೆ.
2. e6ಸ9n' ಫಿಂಟು 2ಫಿಂಟು----- ಲಿಪಿ, ಒಬ್ಬ, ಸಂಖ್ಯೆ, 9ಫಿಂಟು.
 - a) ಸಂಖ್ಯೆ, a
 - b) 18ತೇ
 - c) ಪೆ, ದಿ'ಆ
 - d) ಬಹು2ಂ3.
3. ಕೆಂಪು'ಕೆಂಪು, ಕೆಂಪು'ಕೆಂಪು, ಕೆಂಪು'ಕೆಂಪು?
 - a) e'ಲುಫಿಂ
 - b) ಕೆಂಪು
 - c) &ಂ8
 - d) 2'ಂb.
4. ಸುಕ್ಕ'rod, e86 ಲಿಪಿ ಬ'ದ ಕೆ '.
 - a) ಲಿಪಿಂ8
 - b) d,Sa
 - c) ಕೆಂಪು ಲಿಪಿಂ8
 - d) 2'cಸ ಲಿಪಿಂ8 Are'.
5. ಲಿಪಿಂ8 ಕೆ '88 ಬಹು ಪೆ, ಲಿಪಿ
 - a) 19e ಅಂ8
 - b) ಲಿಪಿ, ಲಿಪಿ ಅಂ8
 - c) 2'ಂ, 9ಕೆ ಅಂ8
 - d) ಕೆ ಲಿಪಿಂ8.

6. 1800 680 བཅོམ་གྱི་དབང་ལྷན་པོ་འཛུགས་པར་

- a) 8. f
b) 99% ರ ಪರಿಣತಿ
- c) ಸೆ 2021 ರ ವರದಿ
d) ಕೆಆರ್‌ಪಿಆರ್‌ ಅಥವಾ ಯು.ಎಸ್.

7. ಪೆಸ್ಟ್ ಲೀನಿಂಗ್ -----ಯ ಸ್ವರೂಪವೇನು?

- a) $\square' r \text{ ಸೆ } \&$ b) 9 ಇ c) ಹಾಳು d) $\square' \text{ ಎಪ } \square$.

8. ಸುತ' ಒಡೆ &ದುವಂ&8rd3 2ಂಜಿ' ರಂ, 3d 2Aಫಂಯ ಅಡುಂ&.

- a) ಬಹುನಾಳಕಾರಿ b) ಕುಸುಮನಾಳಕಾರಿ c) ಚತುರನಾಳಕಾರಿ d) ದ್ವಿನಾಳಕಾರಿ.

9. ಕೃಷಿ ಒಳಗೆ, ಕೃಷಿ ಕೃಷಿ ಕೃಷಿ?

- a) ಆಂಡರೆ, b) ಪಂಪ c) ಬಸವಾಚಾರ್ಯ d) ಹಂಪಿ.

10. d& ಎದ್ದ eಢx.

- a) ಚೌಕ, b) 69 c) ಸೆಂಟಿಮೀಟರ್ d) 30cm.

11. $3\sqrt{5} \times 2\sqrt{3} : 4\sqrt{10} \times 3\sqrt{2}$

- a) ಸ. ೦೭x b) 23ಎಷ, ೦೭x c) ಪೆ6೦೭x d) ಕುಲ೦೭x.

12. □◊or◊a fi◊8ð8Ð ð◊S ◊E2◊ಡುS□d◊?

- a) ತೆ33 b) fiaa c) ರ್ಝaರ್ಝa d) ಘaಘa.

13. 9ಪದಿಸ ಪದದ eಢಫx®'ರ3ಂ?

- a) $\square_{\circ, 0\#}$ b) ಸ. fix c) ಆY d) $2_{\circ\text{ತು.}}$

14. 9⁻. rd. 6ḏ, 3S6 ။ S6c6၀ ။ ။ S□d၀?

- a) 2೦ಡ, b) 2೦ಸ೦೦೦ c) ಭ೦೦, S# d) 2೦೦'Ar3ಹ೦.

15. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದಲ್ಲಿರುವ ಅಕ್ಷರವು _____.

- a) ಭೃಗು ವಂಶಕ b) ಕೃಷ್ಣ c) ಹನುಮಂತ ವಂಶಕ d) ಗುರು.

16. ಕೆಫಿಂಗ್ ಲಿಂಗದ 'ಕೆ' ----- ಹಂಚಿದ ಲಿಂಗದ 'r'
- a) ಕೆಫಿ b) ಹು8 c) ಸಪx d) ಹುತ್
17. ಲಿಂಗದ ಲಿಂಗದ 2fi3 'ಸ' ಸಂ
- a) ಲಿಂಗದ b) ಲಿಂಗದ c) ಲಿಂಗದ d) ಲಿಂಗದ.
18. ಲಿಂಗದ ಲಿಂಗದ A&93 ಆದ, 2 ಲಿಂಗದ ಲಿಂಗದ, ಲಿಂಗದ, ಲಿಂಗದ
- a) ಲಿಂಗದ b) ಕೆಫಿ c) ಲಿಂಗದ d) ಹುತ್, ಲಿಂಗದ.
19. ಲಿಂಗದ 96 ಲಿಂಗದ ----- ಲಿಂಗದ ಲಿಂಗದ 2' ಲಿಂಗದ.
- a) ಲಿಂಗದ b) ಲಿಂಗದ c) ಲಿಂಗದ d) &S.
20. 'ಸಂಘ' Are' ಕೆಫಿ ಕೆಫಿ.
- a) ಲಿಂಗದ, b) 89g c) ಹು' ಲಿಂಗದ d) ಲಿಂಗದ
21. 86v'r ಕೆಫಿ ಲಿಂಗದ, ಸಂಘದ ಲಿಂಗದ, ಲಿಂಗದ.
- a) 2' rT ಲಿಂಗದ b) ಲಿಂಗದ ಲಿಂಗದ ಲಿಂಗದ
- c) 23 ಲಿಂಗದ d) ಲಿಂಗದ ಲಿಂಗದ
22. ತ್ಯ' ಪದದ ಲಿಂಗದ.
- a) 9r6 ಲಿಂಗದ b) ಲಿಂಗದ ಲಿಂಗದ c) ತಂಪ ಲಿಂಗದ d) 9' ಲಿಂಗದ.
23. ಪ, ಲಿಂಗದ 2' ಲಿಂಗದ 2' ಸಂಘ?
- a) 6cಪ b) ಲಿಂಗದ c) ಆ6ಹ d) 9' ಲಿಂಗದ.
24. ಕೆಫಿ ಲಿಂಗದ ಲಿಂಗದ ಲಿಂಗದ ಲಿಂಗದ.
- a) a. i. ಲಿಂಗದ b) ಲಿಂಗದ ಲಿಂಗದ ಲಿಂಗದ
- c) ಲಿಂಗದ, d) ಲಿಂಗದ ಲಿಂಗದ ಲಿಂಗದ.
25. ಲಿಂಗದ ಲಿಂಗದ ಸಂಘದ ಲಿಂಗದ ಲಿಂಗದ S, &fi&G ಲಿಂಗದ ಲಿಂಗದ ಬಂಘದ
- ಲಿಂಗದ, ಲಿಂಗದ ಲಿಂಗದ, ಲಿಂಗದ ಲಿಂಗದ ಲಿಂಗದ ಕೆಫಿ
- a) ಲಿಂಗದ b) ಕೆಫಿ ಸಂಘ

c) ಆ26 ಈ26

d) 2n೦v' Coಬ A&93 ಪSxತೆ. **21KSK37**

26. ಕ&' ವೆdd ಏಢx

- a) ಁA b) □'cr& c) e& d) ಭಞ

27. ಕೆಙ್ಢ ಫಿ'ಠೆಙ್ಢ 8ffi&ಞ Cod ಫ'dS6.

- a) ಕು'ಂಪ□ b) 93ಪೆd6 c) Sಚ3□ಂ66 c) 9v'crಙಂ ಫ'.

28. 192068ಢಂಫಿಪ□6d8D 3□'d e8'rd3----- --.

- a) ಁ&ತ, e8'rd3 b) □ಂon', ಫ e8'rd3
c) ಁ9&rಠ e8'rd3 d) fi,ಂಢಕತx6 e8'rd3.

29. ಕvಂxYಕ ಁ9, d ಢd83 2'ಸ6 c) ----- --.

- a) 2ಂಸ6 c) ಁ9, b) 2ofi&m6 c) ಁ9, c) u'ofi&m6 c) ಁ9, d) ಠೆS□ದ ಁ

30. "ಹ'ಂದಃ' ಉ'c&n' ಢಂಫಿ&GoY, ತ್ಯ'ಠಂದಃ', □'ಁುಂ9ಹ&fi&GoY, dಠ3□' 2ಂ&G
j'rfiಂಲfi&GoY," ad ಁ aS6 Sಚ3ಂಁAj'.

- a) ಬಸSaÂ b) jಂಲ'2ಠ, c) eಕ'22ಂಁj'r9 d) 2ಂ6ಠ, .

31. "ರಂಫid ಫ'crfidಕ'6dn'rಠd n'cಙ್ಢಲಂಞಂಪ□fiæ'ಞ ಆಫಿ6ಂಁd 2ಂ9ಸ&" Cod ಙ್ಢ grS3d 6'ಕೆ'ಠ
ಆddxS3ಂಙ್ಢ ಁ&d ಕ9.

- a) 63| b) 93| c) ಪµ3ಙ d) ಪಂಪ.

32. ಆಠಂ ಁ9, fi&8D ಬ&□'ಠಂಫಿಂS ಫ' ಆಢತ ಫ'ಠಂಁ Hr2ಂx9ಸಂಁA6ಂS ಏಂr'ಁd.

- a) 341v'r ಏಂr'ಁd b) 342v'r ಏಂr'ಁd
c) 343v'r ಏಂr'ಁd d) 344v'r ಏಂr'ಁd.

33. ಪ&ಸ6 ಪ6ಂಪಃ' 2ಂಫic ಪ,ತ,ಠ grS3d ಸಂಁಂ'6fi& 2cಲಕ 23ಂಪ,3 ಏತ6ofi ಪ□'ಠಂS
ಪ&ಪಕ. e'ಠr.

- a) &rಱ b) 9rಱ c) ಠ2x d) ಸಂಸ&ಱ.

34. ಆಞ9ಕ ಕೆಙ್ಢd ಁY ಢಲ 93ಪೆd ಸಂಫi, ಹ fi,ಂಢ.

a) 93ಪದ

b) □ ಂ3ಪದ

c) fi6H0 ²೦ಡು

d) ±೦&ತ, .

35. u'_{θ} ಎೃ, ಪದ್ಧ eಢx

a) $\underline{a}' \text{crA}$ b) $\underline{a}' \text{crfi}$ c) $\text{d} \circ \mathcal{U}^{-1} c, {}^{\circ}'$ d) $\text{M}\overline{\text{e}}\text{s}8.$

36. ಅಂಶ ಪದ ತದ್ಗತವು

a) ಕೆಳಗಿನ 6ರ ಪರಿಣಿತರನ್ನು b) 6ರ ಕೆಳಗಿನ c) 2' d & 1' d) ತತ್ 2 6ರ ಪರಿಣಿತರನ್ನು.

37. ಪೆರ್ಬ' 3□'ಶೃಙ್ಗಿ' C3ೃಙ್ಗಿ□ದೃ ಶೃಙ್ಗಿ □ಫಲಿ -----.

a) ಭಕ್ತಶಾಲ b) ಭಗವಾಶಾಲ c) Sತx2o3ಶಾಲ d) 2x'nಶಾಲ.

38. $\partial_{\alpha} \partial_{\beta} g_{\alpha\beta} = 2 \partial_{\alpha} \partial_{\beta} g_{\alpha\beta}$ e'rb $\partial_{\alpha} \partial_{\beta} g_{\alpha\beta}$

a) v೦ಫಿ b) ee'2೦ಃ c) e' cಡೆS d) n' coಡ. 9v೦ಫಿ.

39. ತಲದಿಹಿಸುಕಂಡ, ಎಂಒ 23ರ & ತವ'ರ 6ಜವ'□ಂ66ಂ.

a) ಕೆಳಗಿನಂತೆ b) ಹಗ್ಗಿನಂತೆ c) ಪದಗಿನಂತೆ d) ಗೋಡೆಯಂತೆ.

40. 2ನಂ' Coಬ A693 ಪSxತ ಪ, ಬಂಧSಯ್ಞ ರ್ತS ಪ, ಿನ ಕದ38od ಆ&□' c&Ñe೦A!.

a) $A \propto \frac{1}{v}$ b) $2\pi \propto v$
c) $A \propto \frac{1}{v^2}$ ಪರಾ d) $A \propto \frac{1}{v^3}$

41. ಸುಂಕೆ ' ' 2' d& ಹ&d n ' c r h ð 8 ð-----ತುಂ9d

a) $\frac{1}{2}A$ b) $C \& G$ c) $\square' cr \&$ d) $\frac{1}{2}A$

42. e' c^a ʔæ ----- dɔ̃ & ʔ Ūrfiɔ̃ S□j'.

a) ཨྲཱུ་ཨྲཱུ་ཨྲཱུ་ b) ཨྲཱུ་ ཨྲཱུ་ c) ཨྲཱུ་ ཨྲཱུ་ d) ཨྲཱུ་

43. $\vec{r} \cdot \vec{s} = 3$, $\vec{r} \cdot \vec{t} = 4$, $\vec{s} \cdot \vec{t} = 5$, $\vec{r} \cdot \vec{u} = 6$, $\vec{s} \cdot \vec{u} = 7$, $\vec{t} \cdot \vec{u} = 8$, $\vec{r} \cdot \vec{v} = 9$, $\vec{s} \cdot \vec{v} = 10$, $\vec{t} \cdot \vec{v} = 11$, $\vec{u} \cdot \vec{v} = 12$, $\vec{r} \cdot \vec{w} = 13$, $\vec{s} \cdot \vec{w} = 14$, $\vec{t} \cdot \vec{w} = 15$, $\vec{u} \cdot \vec{w} = 16$, $\vec{v} \cdot \vec{w} = 17$, $\vec{r} \cdot \vec{x} = 18$, $\vec{s} \cdot \vec{x} = 19$, $\vec{t} \cdot \vec{x} = 20$, $\vec{u} \cdot \vec{x} = 21$, $\vec{v} \cdot \vec{x} = 22$, $\vec{w} \cdot \vec{x} = 23$, $\vec{r} \cdot \vec{y} = 24$, $\vec{s} \cdot \vec{y} = 25$, $\vec{t} \cdot \vec{y} = 26$, $\vec{u} \cdot \vec{y} = 27$, $\vec{v} \cdot \vec{y} = 28$, $\vec{w} \cdot \vec{y} = 29$, $\vec{x} \cdot \vec{y} = 30$, $\vec{r} \cdot \vec{z} = 31$, $\vec{s} \cdot \vec{z} = 32$, $\vec{t} \cdot \vec{z} = 33$, $\vec{u} \cdot \vec{z} = 34$, $\vec{v} \cdot \vec{z} = 35$, $\vec{w} \cdot \vec{z} = 36$, $\vec{x} \cdot \vec{z} = 37$, $\vec{y} \cdot \vec{z} = 38$, $\vec{r} \cdot \vec{a} = 39$, $\vec{s} \cdot \vec{a} = 40$, $\vec{t} \cdot \vec{a} = 41$, $\vec{u} \cdot \vec{a} = 42$, $\vec{v} \cdot \vec{a} = 43$, $\vec{w} \cdot \vec{a} = 44$, $\vec{x} \cdot \vec{a} = 45$, $\vec{y} \cdot \vec{a} = 46$, $\vec{z} \cdot \vec{a} = 47$, $\vec{r} \cdot \vec{b} = 48$, $\vec{s} \cdot \vec{b} = 49$, $\vec{t} \cdot \vec{b} = 50$, $\vec{u} \cdot \vec{b} = 51$, $\vec{v} \cdot \vec{b} = 52$, $\vec{w} \cdot \vec{b} = 53$, $\vec{x} \cdot \vec{b} = 54$, $\vec{y} \cdot \vec{b} = 55$, $\vec{z} \cdot \vec{b} = 56$, $\vec{r} \cdot \vec{c} = 57$, $\vec{s} \cdot \vec{c} = 58$, $\vec{t} \cdot \vec{c} = 59$, $\vec{u} \cdot \vec{c} = 60$, $\vec{v} \cdot \vec{c} = 61$, $\vec{w} \cdot \vec{c} = 62$, $\vec{x} \cdot \vec{c} = 63$, $\vec{y} \cdot \vec{c} = 64$, $\vec{z} \cdot \vec{c} = 65$, $\vec{r} \cdot \vec{d} = 66$, $\vec{s} \cdot \vec{d} = 67$, $\vec{t} \cdot \vec{d} = 68$, $\vec{u} \cdot \vec{d} = 69$, $\vec{v} \cdot \vec{d} = 70$, $\vec{w} \cdot \vec{d} = 71$, $\vec{x} \cdot \vec{d} = 72$, $\vec{y} \cdot \vec{d} = 73$, $\vec{z} \cdot \vec{d} = 74$, $\vec{r} \cdot \vec{e} = 75$, $\vec{s} \cdot \vec{e} = 76$, $\vec{t} \cdot \vec{e} = 77$, $\vec{u} \cdot \vec{e} = 78$, $\vec{v} \cdot \vec{e} = 79$, $\vec{w} \cdot \vec{e} = 80$, $\vec{x} \cdot \vec{e} = 81$, $\vec{y} \cdot \vec{e} = 82$, $\vec{z} \cdot \vec{e} = 83$, $\vec{r} \cdot \vec{f} = 84$, $\vec{s} \cdot \vec{f} = 85$, $\vec{t} \cdot \vec{f} = 86$, $\vec{u} \cdot \vec{f} = 87$, $\vec{v} \cdot \vec{f} = 88$, $\vec{w} \cdot \vec{f} = 89$, $\vec{x} \cdot \vec{f} = 90$, $\vec{y} \cdot \vec{f} = 91$, $\vec{z} \cdot \vec{f} = 92$, $\vec{r} \cdot \vec{g} = 93$, $\vec{s} \cdot \vec{g} = 94$, $\vec{t} \cdot \vec{g} = 95$, $\vec{u} \cdot \vec{g} = 96$, $\vec{v} \cdot \vec{g} = 97$, $\vec{w} \cdot \vec{g} = 98$, $\vec{x} \cdot \vec{g} = 99$, $\vec{y} \cdot \vec{g} = 100$, $\vec{z} \cdot \vec{g} = 101$, $\vec{r} \cdot \vec{h} = 102$, $\vec{s} \cdot \vec{h} = 103$, $\vec{t} \cdot \vec{h} = 104$, $\vec{u} \cdot \vec{h} = 105$, $\vec{v} \cdot \vec{h} = 106$, $\vec{w} \cdot \vec{h} = 107$, $\vec{x} \cdot \vec{h} = 108$, $\vec{y} \cdot \vec{h} = 109$, $\vec{z} \cdot \vec{h} = 110$, $\vec{r} \cdot \vec{i} = 111$, $\vec{s} \cdot \vec{i} = 112$, $\vec{t} \cdot \vec{i} = 113$, $\vec{u} \cdot \vec{i} = 114$, $\vec{v} \cdot \vec{i} = 115$, $\vec{w} \cdot \vec{i} = 116$, $\vec{x} \cdot \vec{i} = 117$, $\vec{y} \cdot \vec{i} = 118$, $\vec{z} \cdot \vec{i} = 119$, $\vec{r} \cdot \vec{j} = 120$, $\vec{s} \cdot \vec{j} = 121$, $\vec{t} \cdot \vec{j} = 122$, $\vec{u} \cdot \vec{j} = 123$, $\vec{v} \cdot \vec{j} = 124$, $\vec{w} \cdot \vec{j} = 125$, $\vec{x} \cdot \vec{j} = 126$, $\vec{y} \cdot \vec{j} = 127$, $\vec{z} \cdot \vec{j} = 128$, $\vec{r} \cdot \vec{k} = 129$, $\vec{s} \cdot \vec{k} = 130$, $\vec{t} \cdot \vec{k} = 131$, $\vec{u} \cdot \vec{k} = 132$, $\vec{v} \cdot \vec{k} = 133$, $\vec{w} \cdot \vec{k} = 134$, $\vec{x} \cdot \vec{k} = 135$, $\vec{y} \cdot \vec{k} = 136$, $\vec{z} \cdot \vec{k} = 137$, $\vec{r} \cdot \vec{l} = 138$, $\vec{s} \cdot \vec{l} = 139$, $\vec{t} \cdot \vec{l} = 140$, $\vec{u} \cdot \vec{l} = 141$, $\vec{v} \cdot \vec{l} = 142$, $\vec{w} \cdot \vec{l} = 143$, $\vec{x} \cdot \vec{l} = 144$, $\vec{y} \cdot \vec{l} = 145$, $\vec{z} \cdot \vec{l} = 146$, $\vec{r} \cdot \vec{m} = 147$, $\vec{s} \cdot \vec{m} = 148$, $\vec{t} \cdot \vec{m} = 149$, $\vec{u} \cdot \vec{m} = 150$, $\vec{v} \cdot \vec{m} = 151$, $\vec{w} \cdot \vec{m} = 152$, $\vec{x} \cdot \vec{m} = 153$, $\vec{y} \cdot \vec{m} = 154$, $\vec{z} \cdot \vec{m} = 155$, $\vec{r} \cdot \vec{n} = 156$, $\vec{s} \cdot \vec{n} = 157$, $\vec{t} \cdot \vec{n} = 158$, $\vec{u} \cdot \vec{n} = 159$, $\vec{v} \cdot \vec{n} = 160$, $\vec{w} \cdot \vec{n} = 161$, $\vec{x} \cdot \vec{n} = 162$, $\vec{y} \cdot \vec{n} = 163$, $\vec{z} \cdot \vec{n} = 164$, $\vec{r} \cdot \vec{o} = 165$, $\vec{s} \cdot \vec{o} = 166$, $\vec{t} \cdot \vec{o} = 167$, $\vec{u} \cdot \vec{o} = 168$, $\vec{v} \cdot \vec{o} = 169$, $\vec{w} \cdot \vec{o} = 170$, $\vec{x} \cdot \vec{o} = 171$, $\vec{y} \cdot \vec{o} = 172$, $\vec{z} \cdot \vec{o} = 173$, $\vec{r} \cdot \vec{p} = 174$, $\vec{s} \cdot \vec{p} = 175$, $\vec{t} \cdot \vec{p} = 176$, $\vec{u} \cdot \vec{p} = 177$, $\vec{v} \cdot \vec{p} = 178$, $\vec{w} \cdot \vec{p} = 179$, $\vec{x} \cdot \vec{p} = 180$, $\vec{y} \cdot \vec{p} = 181$, $\vec{z} \cdot \vec{p} = 182$, $\vec{r} \cdot \vec{q} = 183$, $\vec{s} \cdot \vec{q} = 184$, $\vec{t} \cdot \vec{q} = 185$, $\vec{u} \cdot \vec{q} = 186$, $\vec{v} \cdot \vec{q} = 187$, $\vec{w} \cdot \vec{q} = 188$, $\vec{x} \cdot \vec{q} = 189$, $\vec{y} \cdot \vec{q} = 190$, $\vec{z} \cdot \vec{q} = 191$, $\vec{r} \cdot \vec{r} = 192$, $\vec{s} \cdot \vec{r} = 193$, $\vec{t} \cdot \vec{r} = 194$, $\vec{u} \cdot \vec{r} = 195$, $\vec{v} \cdot \vec{r} = 196$, $\vec{w} \cdot \vec{r} = 197$, $\vec{x} \cdot \vec$

a) ae' cÃr b) m&Ae' cÃr c) d) ae' cÃr.

44. ಕೆಳಕಂಡ $\bar{u}Y$ $\bar{u}d$ ಲ ಸಚ್ಛಿನ್ಯತೆ.

a) ʔ, ʔ' rɦ b) dɔfiŋæ' c) ʔðʔ' ʔʔ'2i d) eʔ'2'ɔj' r9.

45. 'ಸೆ 9ಫಿಂ6ಂ -----ಕೆ86ಲು 26±'ಸಿಫಿಂ
 a) 'ಸೆಸCe' b) 'ಸೆ ಹೆಂ b) 'ಸೆಸೆಂಂಡೆ d) ಹೆ'u'r6ಂ.
46. 'ನ' ಒಮ್ಮೆ, ಪ, 2ಂb ಒಂಢಲ, ಫಿ6ಂತೆ' .
 a) 10 b) 4 c) 6 d) 8.
47. ಕುಂಢು ಒಂರಂ ಪದ, ಸಿಂಞ ಆರ್ದ ಕೆಸ ಸಂಕಲಿ ----- --.
 a) ಫಿ& b) ವಂಢ8re' c) ಸೆ9Arತೆ d) ವಂಕುತಂಹ.
48. 2ನಂವ' ಕೂಬ A & ರೆ8 ಲೆಂಸಂಢಾ6ಂS 9ವಂಫಿ
 a) ನ'ಕೂಡೆ b) ಕೂa9 c) ಲೆ'2ಂ& d) ಒಂಢು93.
49. 'ಸೆಸೆಂ ಬೆ'ರೆ 2ರ83-----ಕೆ'ನ' 2ಕಲಿ ವ'ಲೆ'.
 a) 9ತೆ, b) ಒಡೆ c) 2ಂಢ, a d) ವ'ರಢೆ.
50. ಕೆ, 2, &ಸೂ, ತಕುಬದೇಂಢ ಆ'ಕುಕೆ', 9ಪೆಪ್ಪೆ' ಅS-----6S6 ಹುಂ, ಫಿಂಫಿ&G.
 a) ಸೆ. Co. 9'.rd.6ಢೆ, b) ನಂಢಂ8rಗಿ
 c) ಲೆ'2'ುಂ&G 36ರಂಢ 2ಕಹx d) ರೆಂಸೆಂಢು ಲಿ

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Question Paper Version : B

Third Semester B.E./B.Tech Degree Examination, Jan./Feb. 2023 Samskruthika Kannada

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

ಸಕಲಾಂಗ ಉತ್ತರಿಸಬೇಕಾದ ಪ್ರಶ್ನೆಗಳಿವು :

1. ಸಂಕೇತ '2' ದ ಹೆಸರು ಏನು? -----
- a) ಅಂಕ b) C&G c) 'cr& d) ಭಾಷೆ
2. 'e' ಅಕ್ಷರ ----- ದ ಉಚಿತ ಉಪಯೋಗ.
- a) 'ಎ' b) 'ಆ' c) 'ಇ' d) 'ಒ'
3. ಕೆಳಕಂಡ ಪದಗಳಲ್ಲಿ 'cr' ಅಕ್ಷರ ----- .
- a) 'ಎ' b) 'ಆ' c) 'ಇ' d) 'ಒ'
4. ಕೆಳಕಂಡ ಪದಗಳಲ್ಲಿ 'S' ಅಕ್ಷರ ----- .
- a) 'ಎ' b) 'ಆ' c) 'ಇ' d) 'ಒ'
5. '2' ಅಕ್ಷರ ----- ಕೆಳಕಂಡ ಪದಗಳಲ್ಲಿ.
- a) 'ಎ' b) 'ಆ' c) 'ಇ' d) 'ಒ'

- [illegible]

26. ಕೆಳಗಿನಲ್ಲಿ ಒಂದನ್ನು 'ಕೆ' ----- ಹೊಂದಿರುವುದು
 a) ಕೆಳಗೆ b) ಹೊಂದಿ c) ಸಮಾನ d) ಹೊಂದಿ
27. ಒಂದನ್ನು 2ನೇ 'ಸ' ಸಹಿ
 a) ಎಂದಿ b) ಎಂದಿ c) ಎಂದಿ d) ಎಂದಿ.
28. ಎಂದಿ, 2ನೇ 'ಸ' ಸಹಿ, 2ನೇ 'ಸ' ಸಹಿ, 2ನೇ 'ಸ' ಸಹಿ
 a) ಎಂದಿ b) ಕೆಳಗೆ c) ಹೊಂದಿ d) ಹೊಂದಿ, 2ನೇ 'ಸ' ಸಹಿ.
29. ಒಂದನ್ನು 96ನೇ 'ಸ' ----- ದ್ದು 3ನೇ 'ಸ' ಸಹಿ.
 a) 8ನೇ b) 9ನೇ c) 10ನೇ d) &S.
30. 2ನೇ 'ಸ' ಸಹಿ Are' ಕೆಳಗೆ ಕೆಳಗೆ
 a) u'ROj', b) 89g c) ಕೆಳಗೆ d) 2ನೇ 'ಸ' ಸಹಿ.
31. 99ನೇ 'ಸ' ಸಹಿ 2ನೇ 'ಸ' ಸಹಿ. 2ನೇ 'ಸ' ಸಹಿ ಒಂದನ್ನು ಕೆಳಗೆ ಕೆಳಗೆ?
 a) ಕೆಳಗೆ 'ಸ' ಸಹಿ b) ಬಹು 'ಸ' ಸಹಿ c) ಹೊಂದಿ bಹು' d) i'RSಸಹಿ.
32. ಎಂದಿ 'ಸ' ಸಹಿ 2ನೇ 'ಸ' ಸಹಿ ----- ಒಂದನ್ನು, ಒಂದನ್ನು, 2ನೇ 'ಸ' ಸಹಿ, 2ನೇ 'ಸ' ಸಹಿ.
 a) ಎಂದಿ, a b) 18ನೇ c) ಎಂದಿ, d'f d) ಬಹು 2ನೇ 'ಸ' ಸಹಿ.
33. ಕೆಳಗೆ 'ಸ' ಸಹಿ, 2ನೇ 'ಸ' ಸಹಿ ಎಂದಿ 'ಸ' ಸಹಿ?
 a) ಎಂದಿ b) ಕೆಳಗೆ c) ಹೊಂದಿ d) 2ನೇ 'ಸ' ಸಹಿ.
34. Sಸಹಿ 'ಸ' ಸಹಿ, ಎಂದಿ ಬಹು 'ಸ' ಸಹಿ.
 a) ಹೊಂದಿ b) d,Sa c) ಕೆಳಗೆ ಒಂದನ್ನು d) 2ನೇ 'ಸ' ಸಹಿ ಎಂದಿ Are'.
35. ಹೊಂದಿ ಕೆಳಗೆ 'ಸ' ಸಹಿ ಬಹು 'ಸ' ಸಹಿ, 2ನೇ 'ಸ' ಸಹಿ
 a) 19ನೇ 2ನೇ 'ಸ' ಸಹಿ b) 2ನೇ 'ಸ' ಸಹಿ, 2ನೇ 'ಸ' ಸಹಿ
 c) 2ನೇ 'ಸ' ಸಹಿ, 2ನೇ 'ಸ' ಸಹಿ d) ಕೆಳಗೆ ಎಂದಿ 'ಸ' ಸಹಿ.

36. 1800 680 ಬಾಹ್ಯಾಕಾಶದಲ್ಲಿ ಇರುವ ಕೆಲವು ಆಧಾರಗಳು.
- a) 8. 1 b) 99 ರಿಂದ ಪಂಚತ
c) ಸುಮಾರು 2 ವರ್ಷ d) ಕೆಲವು ಆರಂಭ.
37. ಪದವಿ ಯಾವುದು? a) ರಾಜ್ಯ b) 91 c) ಹಳೆಯ d) ಇವೆಲ್ಲ.
38. ಸುಮಾರು 1800 ವರ್ಷಗಳಿಂದ ಇರುವ 2ನೇ ಅಧ್ಯಾಯವು.
- a) ಬಾಹ್ಯಾಕಾಶ b) ಕುಳಿತುಕೊಳ್ಳುವ c) ಚಲಿಸುವ d) ಇವೆಲ್ಲ.
39. ಕೆಲವು ವಿಷಯಗಳನ್ನು ಕುರಿತು ಕೆಳಕಂಡಿರುವವು.
- a) ಆಂಡರ್ಸನ್, b) ಪಂಪ c) ಬಸವಾಚಾರ್ಯ d) ಹಂಪಿಯ.
40. ದೇಶದ ಹಳೆಯ.
- a) ಚಂದ್ರ, b) 69 c) ಸಂಸ್ಕೃತ d) 3ನೇ.
41. 86 ವರ್ಷಗಳಿಂದ ಇರುವ 2ನೇ ಅಧ್ಯಾಯವು.
- a) 2ನೇ ಅಧ್ಯಾಯ b) 1ನೇ ಅಧ್ಯಾಯ
c) 23ನೇ ಅಧ್ಯಾಯ d) 1ನೇ ಅಧ್ಯಾಯ
42. ತೃತೀಯ ಹಳೆಯ.
- a) 9ನೇ b) 10ನೇ c) 11ನೇ d) 12ನೇ.
43. ಪದವಿ ಯಾವುದು? a) 6ನೇ b) 7ನೇ c) 8ನೇ d) 9ನೇ.
44. ಕೆಲವು ವಿಷಯಗಳನ್ನು ಕುರಿತು ಕೆಳಕಂಡಿರುವವು.
- a) 1ನೇ b) 2ನೇ c) 3ನೇ d) 4ನೇ.
45. ಅನೇಕವು 2ನೇ ಅಧ್ಯಾಯವು.
- a) 1ನೇ b) 2ನೇ c) 3ನೇ d) 4ನೇ.

c) ಆ26 ಈ26

d) 2n೦೩v' Coಬ A&93 ಪSxತೆ **21KSK37**

- [illegible]

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Question Paper Version : C

Third Semester B.E./B.Tech Degree Examination, Jan./Feb. 2023
Samskruthika Kannada

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

ಸಕಲಾಂಗ ಉತ್ತರಿಸುವುದು ಒತ್ತಾಯ :

1. 86v'R ಕೆಳಕಂಡ ಉತ್ತರಗಳಲ್ಲಿ, ಸರಿಯಾದದ್ದನ್ನು ಆಯ್ಕೆಮಾಡಿ.
 - a) 2'RT □ 'cRL
 - b) U, . i'cಡÀ6on'Rnoಡ6o
 - c) 23o ಬರೆಯಿರಿ
 - d) ರೂಪಾಯಿ ೬೫
2. ತ್ಯಾ' ಪದದ ಅರ್ಥ.
 - a) 9R6o
 - b) ಉತ್ತರಿಸುವುದು □ '
 - c) ತಂಪು □
 - d) 9Rಲು.
3. ಪ, 2oದಿ3 2'ಂಡಹ 2'ಸಾ'ರ3o?
 - a) 6cಪ
 - b) 3' R&o
 - c) ಆ6ಹ
 - d) 5o6ಹ.
4. ಕವಯಿತರ ವಿಷಯ ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
 - a) a. i. 3' o'RT
 - b) ಆಲಂಕಾರಿಕ ಕಾವ್ಯವೆಂಬುದು
 - c) u'R0j',
 - d) 23' o'ಕ □ 'RT ಎಂಬುದು.
5. ಅಕ್ಷರವಿನ್ಯಾಸ ಸರಿಪಡಿಸುವುದು □ 'ಸÀ6d8D6o S, &fi&G ಉಳಿಸಿಕೊಂಡು ಬರೆಯುವುದು.

ಉತ್ತರ, ವಿನ್ಯಾಸ, ವಿನ್ಯಾಸ ಹೊಂದಿಸುವುದು ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?

 - a) ರೂಪಾಯಿ ೬೫
 - b) ಕವಯಿತರ ಸಂಪಾದನೆ

c) ಆ26 ಈ26

d) 2n೦೩v' Coಬ A&93 ಪೆSxತ.

6. ಕ&' ಪದದ ಧ್ವನಿ
a) ಅಂಶಾ b) 'CR& c) e& d) ಭರ್ಷ
7. ಕೆಳಗಿನ ಕ್ರಿಯಾತ್ಮಕ ಕೋಡ್‌ಗಳಲ್ಲಿ ಕೆಳಗಿನ
a) ಕುಂಪಿತ b) ಗ್ರಹದಂತಿ c) ಸಚ್ಚಿಂತನಾ d) ಗ್ರಹೀತೃ ಕ್ರಿಯಾತ್ಮಕ.
8. 1920ರಲ್ಲಿ ಬಳಸಿದ ಮೊದಲಿನಿಂದಲೂ ಬಳಸಲಾಗುತ್ತಿರುವ ಸಂಕೇತವು -----
a) ಅಕ್ಷರ, ಸಂಕೇತ b) ಅಕ್ಷರ, ಸಂಕೇತ
c) ಅಕ್ಷರ, ಸಂಕೇತ d) ಅಕ್ಷರ, ಸಂಕೇತ
9. ಕೆಳಗಿನ ಯಾವುದು ಸರಿಯಾದ ಸಂಕೇತವು -----
a) ಅಕ್ಷರ, ಸಂಕೇತ b) ಅಕ್ಷರ, ಸಂಕೇತ c) ಅಕ್ಷರ, ಸಂಕೇತ d) ಅಕ್ಷರ, ಸಂಕೇತ
10. "ಹೆಚ್‌ಎಲ್‌ಎಲ್‌" ಎಂಬುದು ಒಂದು ಸಂಕೇತವು, ಅದು "ಹೆಚ್‌ಎಲ್‌ಎಲ್‌" ಎಂಬುದು, ಅದು "ಹೆಚ್‌ಎಲ್‌ಎಲ್‌" ಎಂಬುದು, ಅದು "ಹೆಚ್‌ಎಲ್‌ಎಲ್‌" ಎಂಬುದು.
a) ಬಸಸಾ b) ಜಿಎಲ್‌ಎಲ್‌ c) ಎಚ್‌ಎಲ್‌ಎಲ್‌ d) ಎಚ್‌ಎಲ್‌ಎಲ್‌
11. ಸಂಕೇತವು '2' ದಿಂದ ಹಿಂದಿನಿಂದಲೂ ಬಳಸಲಾಗುತ್ತಿರುವ ಸಂಕೇತವು -----
a) ಅಂಶಾ b) C&G c) 'CR& d) ಭರ್ಷ
12. e'c'a'fæ-----ದಂತಿ ಉಪಯೋಗಿಸಿ.
a) ಎಚ್‌ಎಲ್‌ಎಲ್‌ b) ಎಚ್‌ಎಲ್‌ಎಲ್‌ c) ಹೆಚ್‌ಎಲ್‌ಎಲ್‌ d) Bod'.
13. ಕೆಳಗಿನ ಕೆಳಗಿನ ಅಕ್ಷರವು 'CR&' -----
a) ಅಕ್ಷರ, ಸಂಕೇತ b) ಅಕ್ಷರ, ಸಂಕೇತ c) ಅಕ್ಷರ, ಸಂಕೇತ d) ಅಕ್ಷರ, ಸಂಕೇತ
14. ಕೆಳಗಿನ ಯಾವುದು ಸರಿಯಾದ ಸಂಕೇತವು -----
a) ಹೆಚ್‌ಎಲ್‌ಎಲ್‌ b) ದಂತಿ, ಸಂಕೇತ c) ಅಕ್ಷರ, ಸಂಕೇತ d) ಎಚ್‌ಎಲ್‌ಎಲ್‌.
15. 'ಸೆ' ಗಳಲ್ಲಿ -----ಕೆಳಗಿನ 26 ಸಂಕೇತಗಳಿವೆ
a) 'ಸೆ' b) 'ಸೆ' ಹೆಚ್‌ಎಲ್‌ಎಲ್‌ c) 'ಸೆ' ಅಕ್ಷರ, ಸಂಕೇತ d) ಹೆಚ್‌ಎಲ್‌ಎಲ್‌.

25. ಉತ್ಪತ್ತಿ, ಪದದ ಉದಾಹರಣೆ

- a) ಅ'cra b) ಅ'crfi c) ದಂ'c, ' d) ಮೆ8.

26. ಅಂಶದ ಪದದ ತದ್ವಿಧಿ

- a) ಕೌಂಟಿ 6ನೇ b) 6ನೇ ಸ c) 2' d& ' d) ತತ್ತ್ವ 2 6ನೇ ಸಲಿ.

27. ಪರಿಣಿತರ 3ನೇ ಸ್ಥಾನದಿಂದ ಕೆಳಗೆ ಬಂದಿರುವವರಲ್ಲಿ ಯಾರು ಸರಿಯಾದವರು? -----

- a) ಭಕ್ತರ b) ಭಕ್ತರ c) ಸತ್ಯ 2ನೇ ಸ್ಥಾನ d) 2ನೇ ಸ್ಥಾನ.

28. ಕೆಳಗೆ ನೀಡಲಾಗಿರುವವರಲ್ಲಿ ಯಾರು ಸರಿಯಾದವರು?

- a) ವರ್ತಿ b) ಎ'2ನೇ c) ಎ'ಕಡೆ d) ನ'ಕಡೆ. 9ನೇ ಸಲಿ.

29. ತಲೆಗಿಳಿಸಿಕೊಂಡು, ಎಂಟು 23ನೇ ಸ್ಥಾನದಿಂದ 6ನೇ ಸ್ಥಾನಕ್ಕೆ ಬಂದಿರುವವರಲ್ಲಿ ಯಾರು ಸರಿಯಾದವರು?

- a) ಕೆಳಗೆ ಬಂದವರು b) ಹೇಗೆ ಬಂದವರು c) ಪರಿಣಿತರ 6ನೇ ಸ್ಥಾನದಿಂದ ಬಂದವರು d) ಬಂದವರು.

30. 2ನೇ ಸ್ಥಾನದಿಂದ 6ನೇ ಸ್ಥಾನಕ್ಕೆ ಬಂದವರಲ್ಲಿ ಯಾರು ಸರಿಯಾದವರು? ಅಂಶದ ಕೆಳಗೆ ಬಂದಿರುವವರಲ್ಲಿ ಯಾರು ಸರಿಯಾದವರು?

- a) A&93 b) 2ನೇ ಸ್ಥಾನದಿಂದ c) A&93 ವರ್ತಿ 8ನೇ ಸ್ಥಾನದಿಂದ d) ಆ'ಕಡೆ.

31. 3ನೇ ಸ್ಥಾನದಿಂದ 6ನೇ ಸ್ಥಾನಕ್ಕೆ ಬಂದವರಲ್ಲಿ ಯಾರು ಸರಿಯಾದವರು?

- a) ಸ. 6ನೇ b) 23ನೇ ಸ್ಥಾನದಿಂದ, 6ನೇ c) ಪರಿಣಿತರ 6ನೇ ಸ್ಥಾನದಿಂದ d) ಕೆಳಗೆ ಬಂದವರು.

32. ಕೆಳಗೆ ನೀಡಲಾಗಿರುವವರಲ್ಲಿ ಯಾರು ಸರಿಯಾದವರು?

- a) ತತ್ತ್ವ b) fiaa c) ಝಾಝಾ d) ಝಾಝಾ.

33. 9ನೇ ಸ್ಥಾನದಿಂದ 6ನೇ ಸ್ಥಾನಕ್ಕೆ ಬಂದವರಲ್ಲಿ ಯಾರು ಸರಿಯಾದವರು?

- a) ಕೆಳಗೆ ಬಂದವರು b) ಸ. fix c) ಆ' d) 2ನೇ ಸ್ಥಾನ.

34. 9ನೇ ಸ್ಥಾನದಿಂದ 6ನೇ ಸ್ಥಾನಕ್ಕೆ ಬಂದವರಲ್ಲಿ ಯಾರು ಸರಿಯಾದವರು?

- a) 2ನೇ ಸ್ಥಾನದಿಂದ, b) 2ನೇ ಸ್ಥಾನದಿಂದ c) ಭಕ್ತರ, ಸ d) 2ನೇ ಸ್ಥಾನದಿಂದ.

35. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಭೂಮಿ b) ಕೆಳಕಂಡು c) ಹಿರಿಮೆ d) ಗುರುತು.
36. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಕೆಳಕಂಡು b) ಹಿರಿಮೆ c) ಸುಖ d) ಹುಡುಗ.
37. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಕೆಳಕಂಡು b) ಹಿರಿಮೆ c) ಸುಖ d) ಹುಡುಗ.
38. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಕೆಳಕಂಡು b) ಹಿರಿಮೆ c) ಸುಖ d) ಹುಡುಗ.
39. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಕೆಳಕಂಡು b) ಹಿರಿಮೆ c) ಸುಖ d) ಹುಡುಗ.
40. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಕೆಳಕಂಡು b) ಹಿರಿಮೆ c) ಸುಖ d) ಹುಡುಗ.
41. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಕೆಳಕಂಡು b) ಹಿರಿಮೆ c) ಸುಖ d) ಹುಡುಗ.
42. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಕೆಳಕಂಡು b) ಹಿರಿಮೆ c) ಸುಖ d) ಹುಡುಗ.
43. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಕೆಳಕಂಡು b) ಹಿರಿಮೆ c) ಸುಖ d) ಹುಡುಗ.
44. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಕೆಳಕಂಡು b) ಹಿರಿಮೆ c) ಸುಖ d) ಹುಡುಗ.
45. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ 2ನೇ ಸ್ಥಾನದ ಅಕ್ಷರವು ----- ಆಗಿರುತ್ತದೆ.
- a) ಕೆಳಕಂಡು b) ಹಿರಿಮೆ c) ಸುಖ d) ಹುಡುಗ.

c) 2019, 9th ao908

d) 30th ao908.

21KSK37

46. 1800 680 ಬಾಷೆಗೆ 'ಗ್ರೇ'ದ ಕೆಳಗೆ ಆದಾಸು.

a) 8. 1'

b) 99ರಲೆರೆ ಪೆಂ8ತ

c) ಸು ಂ21 2vcr

d) ಕBRಪ35 ಆರ್ UR8.

47. ಪೆS&d ಲೆ'n'-----aY,S6ಂ ರೆಂ6ಂ?

a) □'Rಸ&

b) 91

c) ಹಗಂ

d) □'ಂಪ□.

48. ಸುಫ'ಫ& &ದಂvಂ&8rd3 2ಂoj' ಡಂ,3d 2Aಫodಂ aಡುಂ&.

a) ಬಾ'ನಂ6&

b) ಕುouಂ6&

c) ಚ2ಂ6&

d) &ದಂvಂ&.

49. ಕೃತಿ6 □ಂs, ಕೃತಿ ಕತೃ ರೆಂ6ಂ?

a) ಆಂಡರೆ,

b) ಪಂಪ

c) ಬಸSaÂ

d) ಹಂಪಂ.

50. d& ಪdd eಫx.

a) ಚod,

b) 69

c) ಸcಡx

d) 3ಫತ,.

* * * * *

USN

[illegible]

Question Paper Version : D

Third Semester B.E./B.Tech Degree Examination, Jan./Feb. 2023
Samskruthika Kannada

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

ಸರ್ಕಾರದ ಉತ್ಪಾದನೆಯು 29% ಉತ್ಪಾದನೆ :

- [illegible]

5. u'ೞಿಷ9, ಪದ್ಧ eಧx
 a) ಅ'cRA b) ಅ'cRfi c) ದಂ'c, ' d) Mಷ8.
6. ಂಸ ಪದ್ಧ ತದS 6ಪ
 a) ಕಾಂ 6ಪ b) 6ಕ'ಸ c) 2' d&□' d) ತತ, 2 6ಪSಲಿ.
7. ಪ&rb' 3□'ಶಂHĀi' C3ಯCS□ದಂ ಶಂS □ಲಿ ----.
 a) ಭತ□ಲಿ b) ಭ9ಷe□ಲಿ c) Sತx2ಂ3□ಲಿ d) 2ಁ'ನಲಿ.
8. ಶಂS 9ವಂofid 9H, Rಕ6a e'RBಕ6 2Hಕ6
 a) vಂfi b) ee'2ಂ& c) e'cಡS d) n'coಡ. 9ವಂofi.
9. ತಲಿಶಿಸ8ಂಕಂಡ, eಂ&G 23'R &Rತxv'ಶ 6ಜವ'□ಂ66ಂ.
 a) ಕ3ಕಿಂಸ6ಂ b) ಹ&ಿಂಸ6ಂ c) ಪ□6od6iಂಸ6ಂ d) iಂಸ6ಂ.
10. 2ನಂv' Coಬ A693 ಪSxತ ಪ, ಬಂಫS3ಯC ಶಂS ಪ, ಂಸ ಕಥ38od ಅ&□'c&ÑeಂAi'.
 a) A&93 b) 2ನಂv'
 c) A&93 vಂ8n' ಪರಾ d) ಆv'□ಂಡು.
11. 99ಶ3fi6 □ಲಿದ8ಶ 92x ಸ23. ಶS3ಯC □ಂಪಂ8d ಅಂ9 ಶಂ6ಂ?
 a) ಕೞಷĀi'RSಅಂಶ b) ಬುಕ'ಅಂಶ c) ಹಕ'ಬುಕ' d) i'RSಅಂಶ.
12. e6ಸ9n' fioಡು 2fiಂಶಂdā'----- □'cಯ, ಒಬ, ಸಂ, a 9g, dಂAಂಯ.
 a) ಸಂ, a b) f8ತೆ' c) ಪ, d'Ā d) ಬಹು2ಂ3.
13. ಕಂಖYಕದ ಅಂ9, ಫಂ'ಶೆಂಯC ಏv'odಂ ecrĀi'?'
 a) e'ಲುಫಿಂ b) ಕ3Cಡ c) &ಂ8 d) 2ಅಂb.
14. Sಸು¢'rod, eS6ಂ ಬಾ'd ಕ '.
 a) ಶಂನಂಶ b) dSa c) ಕಂಯಡು □ಂಪಂ8a d) 2'cಸ ಂಶ3 Are'.

15. ಕೆಳಕಂಡ ಕೆಲಸಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) 19ನೇ ಅಕ್ಟೋಬರ್ b) 19ನೇ, 20ನೇ ಅಕ್ಟೋಬರ್
c) 20ನೇ, 19ನೇ ಅಕ್ಟೋಬರ್ d) ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದೂ ಸರಿಯಲ್ಲ.
16. 1800 ರಿಂದ 1850 ರ ವರೆಗೆ ಕೆಳಕಂಡ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) 8. 1 b) 99ನೇ ಅಕ್ಟೋಬರ್ ಪಂಚಮಿ
c) ಸಹಿ 20ನೇ 2ನೇ ವರ್ಷ d) ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದೂ ಸರಿಯಲ್ಲ.
17. ಪದವಿ 'ನ' -----ಯ ಸಹಿ ಹೇಗೆ?
- a) 19ನೇ b) 9ನೇ c) ಹೇಗೆ d) 19ನೇ.
18. ಸಹಿ 'ನ' & ದಯವಿಟ್ಟು 2ನೇ, 3ನೇ 2ನೇಯ ಅಕ್ಷರವನ್ನು.
- a) ಬಹುನೇ b) ಕುಂಠನೇ c) ಚಕ್ರವೇ d) & ದಯವಿಟ್ಟು.
19. ಕೆಳಕಂಡ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂತರ, b) ಪಂಪ c) ಬಸವಾಚಾರ್ಯ d) ಹಂಪವ.
20. ದೇವರ ಪದವಿ ಯಾವುದು.
- a) ಚಕ್ರ, b) 69 c) ಸಹಿ d) 3ನೇ, .
21. ಸಹಿ 'ನ' 2ನೇ ದೇವರ ಹೇಗೆ ನೇಮಕಗೊಂಡಿತು-----ತುಂಬಿ.
- a) 19ನೇ b) C&G c) 19ನೇ d) ಭಕ್ತ
22. 'ನ' & ದಯವಿಟ್ಟು 2ನೇಯ ಅಕ್ಷರವನ್ನು.
- a) 2ನೇ ದೇವರ b) 2ನೇ ತೇವರ c) ಕುಂದೇವರ d) Bodೇವರ.
23. ಕೆಳಕಂಡ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) 19ನೇಯ ಅಕ್ಷರವನ್ನು b) 19ನೇಯ ಅಕ್ಷರವನ್ನು c) 19ನೇಯ ಅಕ್ಷರವನ್ನು d) 19ನೇಯ ಅಕ್ಷರವನ್ನು
24. ಕೆಳಕಂಡ ಯಾವುದು ಸರಿಯಾದದ್ದು.
- a) 19ನೇ b) ದಯವಿಟ್ಟು c) ಅಕ್ಷರವನ್ನು d) 19ನೇಯ ಅಕ್ಷರವನ್ನು.

25. 'ಸೆ 9ಫಿಂ6ಂ -----ಕೆ86ಲು 26±'cಬಫಿಂ
 a) 'ಸೆCe' b) 'ಸೆ ಹೇಂ c) 'ಸೆಂಂಡ d) ಹೇ'u'R6ಂ.
26. 'ನ' ಒಷು, ಪ, 2ಂb ಂೌದಲ, ಫಿಂ6ಂತೆ' .
 a) 10 b) 4 c) 6 d) 8.
27. ಕು6ಂಡು ಂೌಂೌಂ ಪದ, S3ಂೃ ಆರೊ ಕೆS3 ಸಂಕಲ3 ----- --.
 a) fi& b) ವಂಎd8Re' c) ಸ9ARತ d) ವಂಕುತಂH.
28. 2ನಂv' Coಬ A & ರ 8 ಲಂಸಂೌಂA6ಂS 9ವಂofಿ
 a) n'coಡ b) ಕುa9 c) ee'2ಂ& d) ಂೌಡು93.
29. 'ೌ6ತೆS ಒ' , ರ 2R83-----ಕೆ'n' 2cಲ v'e' .
 a) 9ತ, b) ಒ& c) 2ಂd, a d) v'Rರೃ.
30. ಕ, 2, &ಸು, ತಕxಬದ&ೌಂಎ ಆ'e'crಚv', 9ಪ&ಪ್ಪ'e' aS-----6S6 ಹುಯ, ಫಿಂafಿ&G.
 a) ಸೆ. Co. 9'. Rd. 6ರ, b) ನಂೌಂ8Rಗ್
 c) e&'2'ುಂ, &G 36ರಂಹ 2cHx d) ರಂSಂdಂ ಲಿ
31. 86v'R ಕೆ3ೃಡ ±ಂ&ತ, ಸ2ರೂ3d eೃ, ಸಂೌಂAd&S6ಂ.
 a) 2'RT ಂ'cRL b) ು, . 'cಡ&6on'Rಂೌಡ6ಂ
 c) 23ಂ ಬರಂೌ6 d) ರಂSಂdಂ ಲಿ
32. ತ್ಯಂ' ಪದ್ಧ eಫ್ಫx.
 a) 9R6ಂ b) ಂೌರಂ&ಂ' c) ತಂಪಂ d) 9ರಲು.
33. ಪ, 2ಂಎd3 'ರಡಹರ 'ಸಾ'R3ಂ?
 a) 6cಪ b) ಅ'R&ಂ c) ಆ6H d) 'ೌ6H.
34. ಕಂೌxYಕ ಫಿತಂ'ೌಭಸ ಕೃಹರೇಂೃ 69ರS6ಂ.
 a) a. i. ಅ'ೌT b) ಆಲc6ಂ *'ಂಕYೌಂರೇಂ
 c) u'R0j', d) 2& *'ಂಕಂ'RT eರ, onಂ&.

[illegible]

a) ರೆಂಟೆ
b) ಕೆಲಸದ ಸಂಪನ್ಮೂಲ
c) ಆ26 ಈ26
d) 2ನೇ ವರ್ಷದ Coಬ A&93 ಪರಿಷತ್.

36. ಕೆಒ' ಪದಧರ್ಮ

a) $\frac{1}{2}A$ b) $\frac{1}{2}CR$ c) eR d) $\frac{1}{2}A$

37. ಕೆಳಕಂಡ ಕ್ರಮದಲ್ಲಿ ಕೆಳಕಂಡ ಕೋಡ್‌ಗಳನ್ನು ಕೊಡು.

a) ಕು®'ಂಪ□ b) 93ಪೆd6ು c) Sಚೆ3□ಾ66ು d) 9v' ರಗ್ತಾ ತಾ®'.

38. 192068Dvೌಫಿವೃ6d8D 3□'d e8®'Rd3 ----- --.

a) $\pm 0.03, e8^{\circ} \text{ Rd3}$ b) $\square_{\text{on}}, i^{\circ} e8^{\circ} \text{ Rd3}$
c) $\pm 0.09 \text{ Rd3}, e8^{\circ} \text{ Rd3}$ d) $\text{fi}, \text{on}, \text{d}, \text{Rd3}, e8^{\circ} \text{ Rd3}$.

39. ಕವಂxYಕ ಁಂ9, d ūd83 2' ಸ6ಂ ----- --.

a) 2000, b) 2001, c) 2002, d) 2003

40. "ಹೌದ್ ಉಂ' ಎನ್' ಬಂತೆಫಿಗೊಯ್, ತ್ಯು' ಹೌದ್', □' ಅ' ಯಾಹ್‌ನಿಗೊಯ್, ದಶೆ' ವಾಗ್
ಇರ್‌ಫಿಯಲಿಗೊಯ್, " ಆಯ್ ಆಸೆ ಸಚೆವಾಗಿ'.

a) ಬಸSaÂ b) i_ಾl'2ð, c) eಕ'2²_ಾj'_R9 d) 2_ಾ6ð,.

41. $3f \rightarrow 3d$ ಸಹಜ $2x : 3A \rightarrow 3d$

a) ಸ. ೪2x b) 23ಂಪ, ೪2x c) ಪ6೪2x d) ಕುಲ೪2x.

42. ☐ ಎಂದಾ ವಿಷಯವು ಸರಿ ☒ ಅಲ್ಲವೇ?

a) ತ33 b) fiaa c) ಝaಝa d) ಘaಘa.

43. 9ಪೆಢS ಪೆdd eಢx®'R3ು?

a) $\square_{\infty, 0\#}$ b) $\vec{s}.fix$ c) ΩY d) $2_{\infty\#}$.

44. 9-'. Rd. 68, 3S6 ತS6c6ಂ ರೆಂS□dಂ?
- a) 2ಂಡೆ, b) 2್ಯಸ6ಂ c) ಭೆಂ, S# d) 2್ಯ'À3ಹೆ.
45. ಕೆ3ಕೆಂಸ6 2cಲ 2'ಸ6ಂ-----.
- a) ಭ62aÂ vಂರೆಕ b) ಕೆ3ಕಪ್ c) H2ಲಪ್, vಂರೆಕ d) 9R6ಪ್.
46. ಕ&Ñfiog □ಂಡ 2'ಕ'□' -----Hಂಬುd 2ಂಂಬುj'R
- a) ಕ&Ñ b) ಹು8 c) ಸಪx d) ಹುತ್
47. □ಂಫಂ±ಂಬ6 2fi3 2'ಸ6ಂ
- a) e±ಂಢo b) edR c) eಬುÅe d) a±ಂಲ&e.
48. eಂ&fiಂಪ್, d A&93 ಆd, 2 ಂe'ರೆ 2ಂ&'c, Rಂಎಂ, ರೆ6ಂ
- a) a±ಂಲ&e b) ಕ&ರೆ c) ರೆಂಕು d) ಹುಚೆಪ್, 2ಂಸÀ6.
49. □ಂಕ8D 96ತvಂd□'----- ddx3*ಂdಡc 2ಃ'ಠು'Rಕು.
- a) 8ofi b) 9ofi2 c) fiಂ6ಂ d) &S.
50. 2'cಸುಂಲ3 Are' ಕS3d ಕ9.
- a) u'Roj', b) 89ê c) ಕು®'ಂಪ□ d) 2ಂಲÀ.

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